Digital Image Processing:

3 Image Enhancement---Spatial domain[1]

CHENGLU WEN

Department of Cognitive Science

clwen@xmu.edu.cn

LABS!

Day: W6, Tuesday

Time: 14:30 – 18:20pm

Room: Wenxuan building 4#B404,405

Matlab, Matlab Image Processing Toolbox

Department of Cognitive Science
Xiamen University

Contents

Over the next few lectures we will look at image enhancement techniques working in the spatial domain:

1. What is image enhancement?
2. Intensity transformation enhancement
3. Histogram processing
4. Neighbourhood operations(next class)

Spatial domain?

Recall: Grey Levels

In our class, image grey level values we have said are in the range [0, 255]

- Where 0 is black and 255 is white.

Specific reason we use this range

- The range [0,255] stems from display technologies

For many of the image processing operations in this lecture grey levels are assumed to be given in the range [0.0, 1.0]

1 What is Image Enhancement?

Image enhancement——

the process of making images more suitable for a specific application

The reasons for doing this include:

- Highlighting interesting detail in images
- Removing noise from images
- Making images more visually appealing

Spatial & Frequency Domains

There are two broad categories of image enhancement techniques

(a) Spatial domain techniques

- Direct manipulation of image pixels: *Image Plane*

(b) Transform domain techniques

- Manipulation of Fourier transform or wavelet transform of an image

This lecture we will concentrate on techniques that operate in the spatial domain.

Image enhancement

– Spatial domain methods

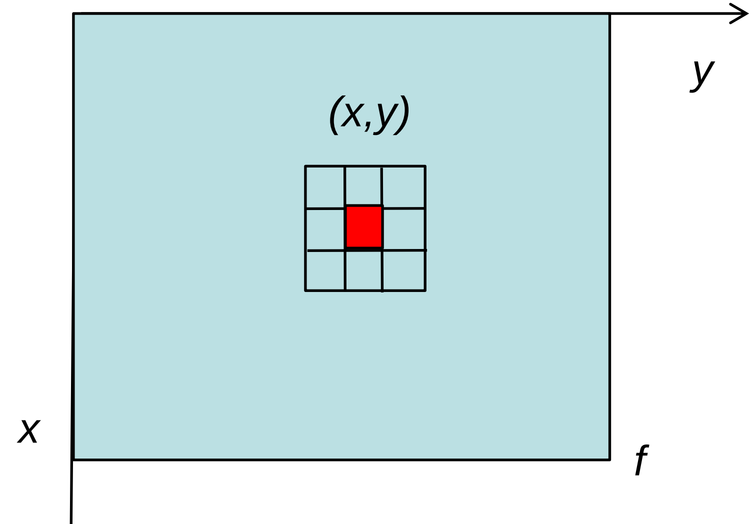
(1) Neighborhood-based(Mask-based) methods - *spatial filter*

(2) Point-based methods - *intensity transformations*

– *Negative*
– *Log transformation*
– *Power-law*
– *Contrast stretching*
– *Gray-level slicing*
– *Histogram equalization*
– *Averaging*

- Frequency domain methods

Basics of spatial domain processing



$$g(x,y) = T[f(x,y)]$$

- *f(x,y): Input image*
- *g(x,y)=Output image*
- *T: operator on f defined over a* ***neighborhood*** *of a pixel x,y*
  - ✓ standard approach: T is applied to a sub-image centred at (x,y).
  - ✓ sub-image is called mask(kernel, filter, template, window)
  - ✓ mask processing or filtering
  - ✓ T can operate on a set of images

Department of Cognitive Science
Xiamen University

2 Intensity transformations

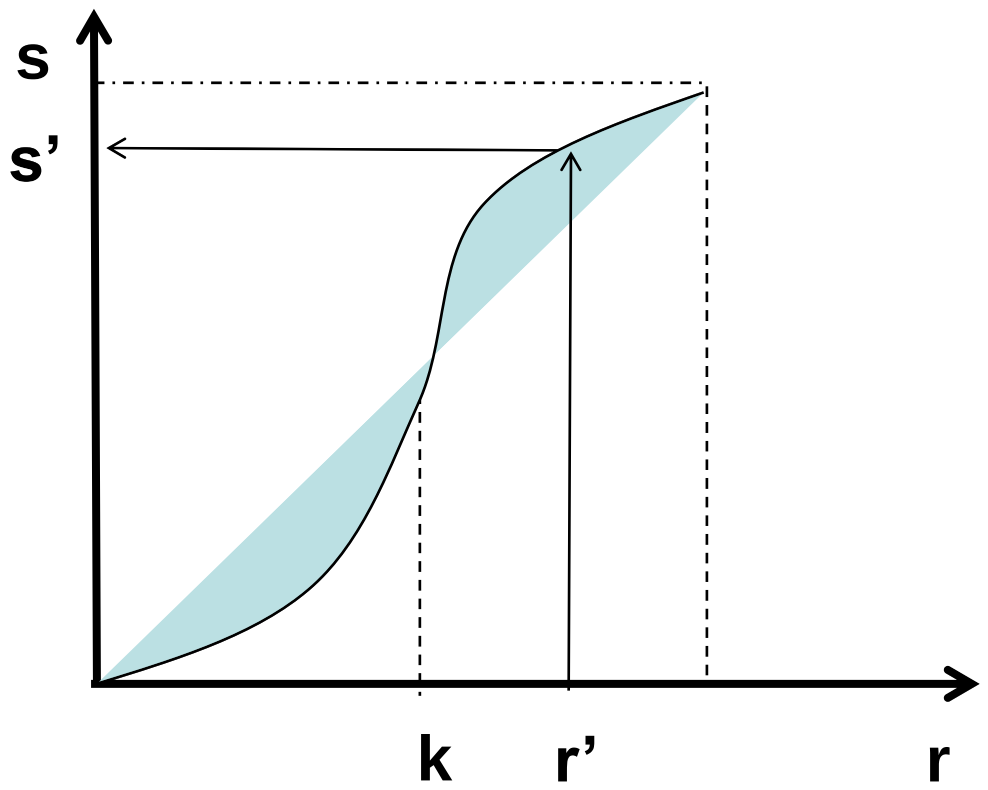
Point processing——

The simplest kind of range transformations are these independent of position *x,y*:

$$g(x,y) = T[f(x,y)]$$

⇩

$$s=T(r)$$



2 Intensity transformations

$$g(x,y) = T[f(x,y)]$$

$$s=T(r)$$

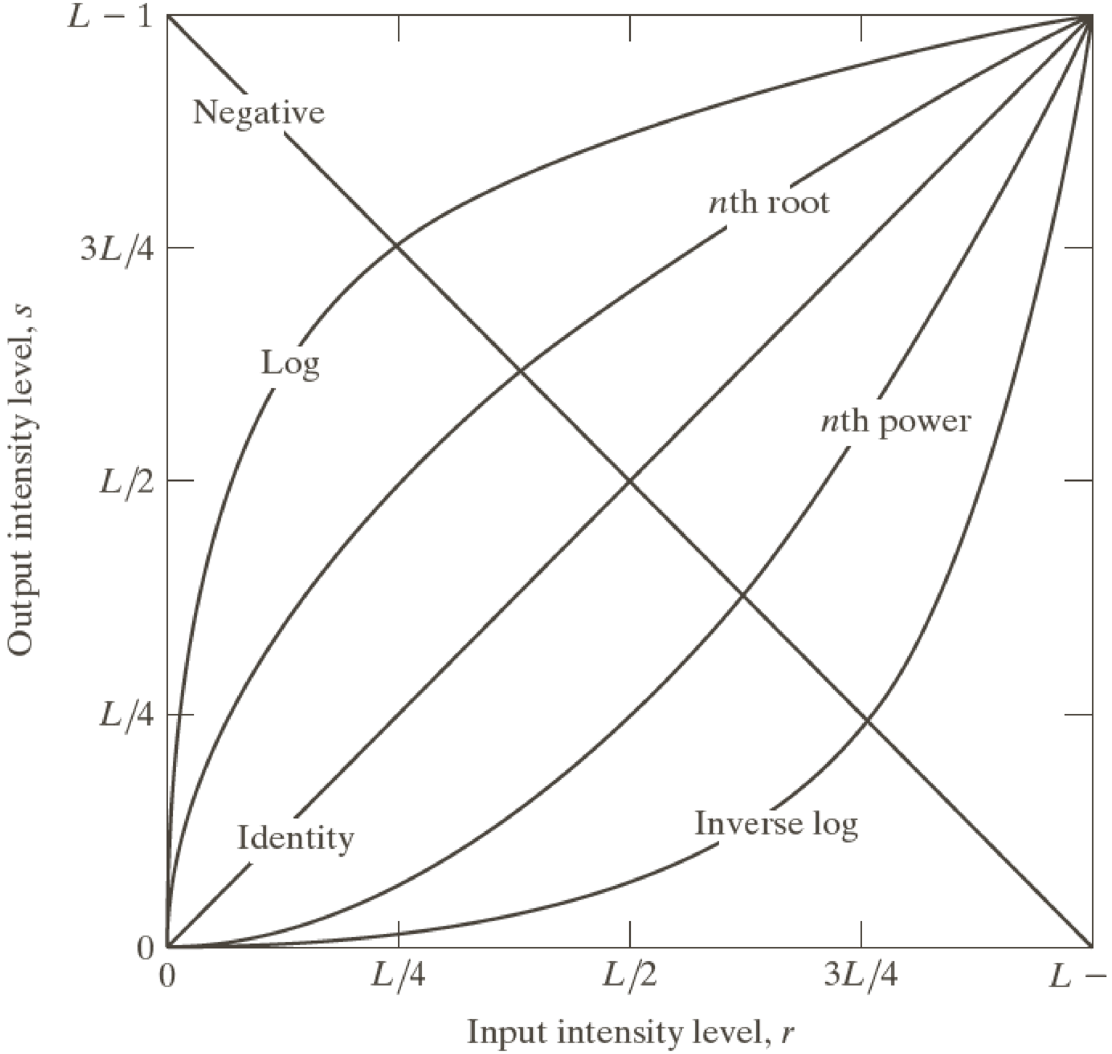
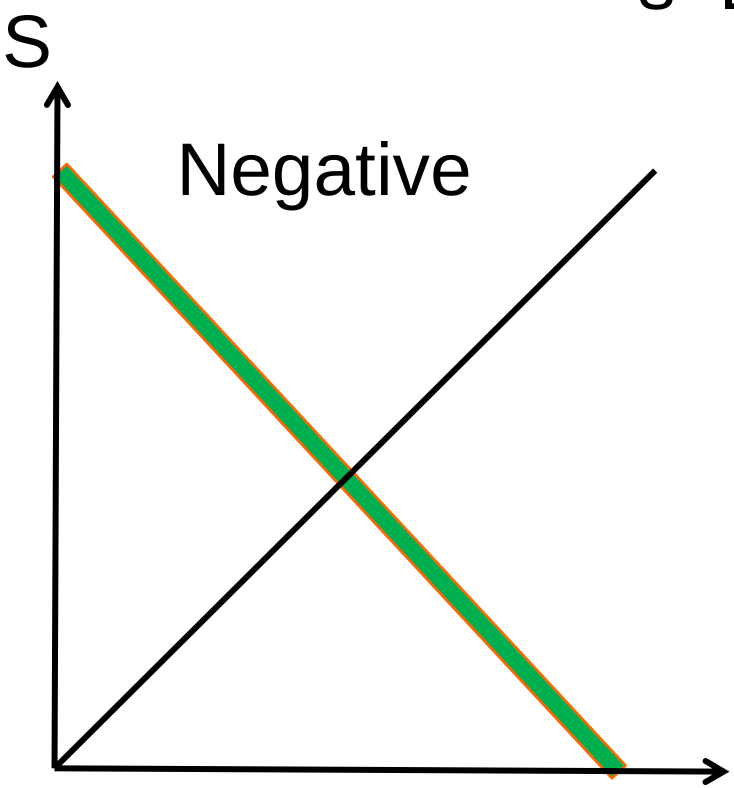


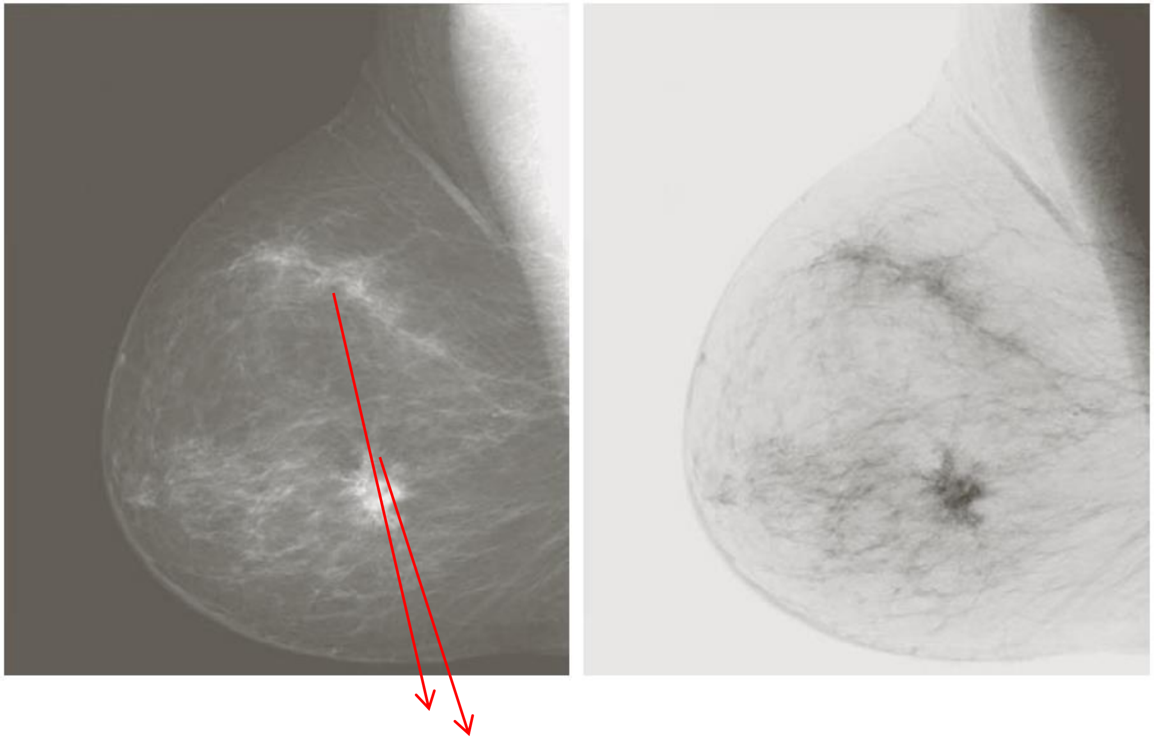
FIGURE 3.3 Some basic intensity transformation functions. All curves were scaled to fit in the range shown.

2 Intensity transformations

(1) Image Negatives

s=L-1-r,   (L intensity level range)

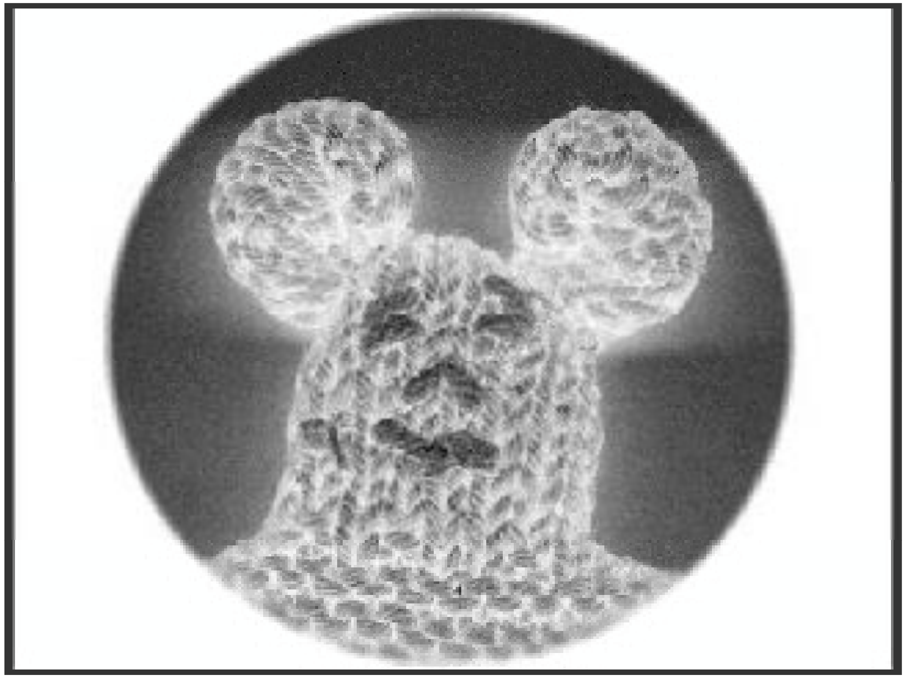




Interest area

2 Intensity transformations

(1) Image Negatives



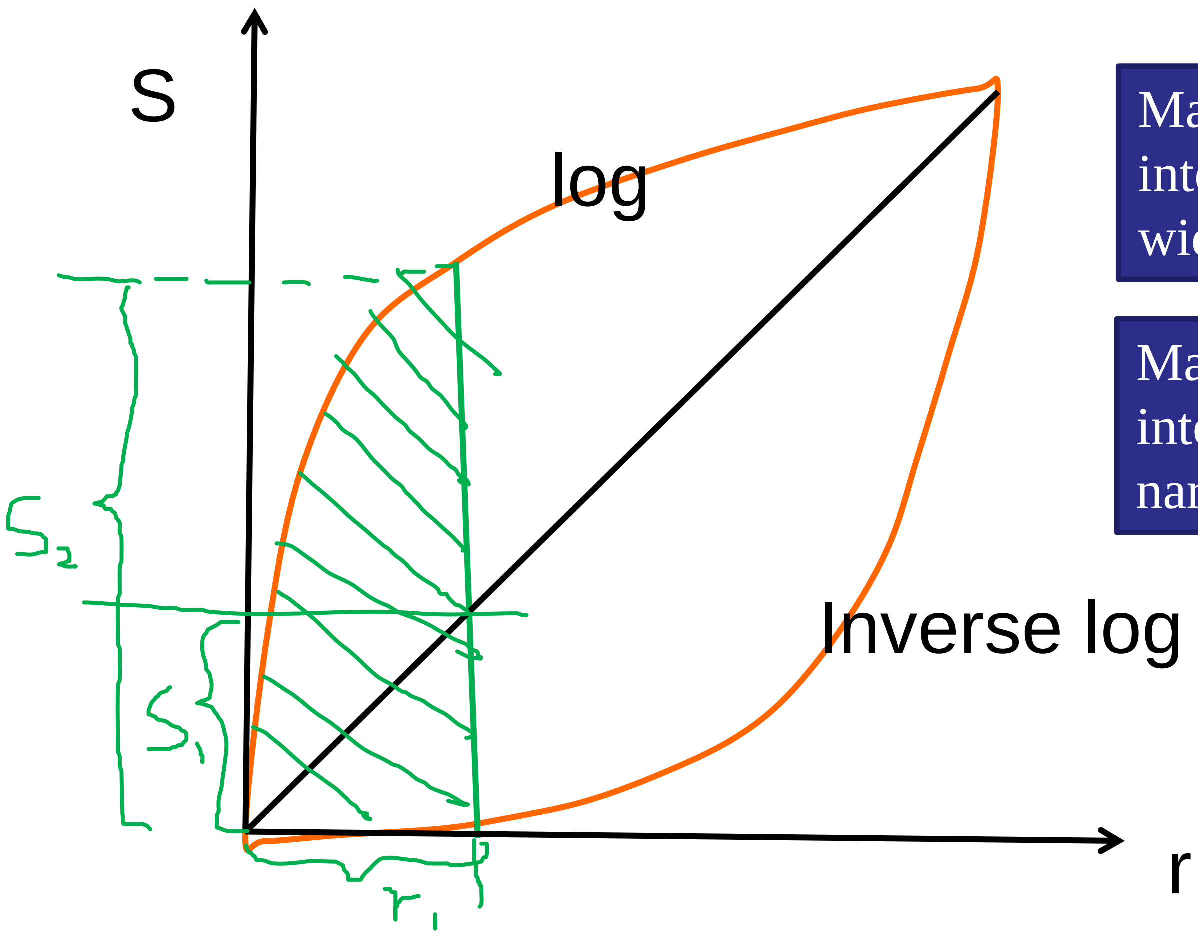
Suited for:
Enhancing white or gray detail embedded in dark regions of an image with a black areas dominant in size.

Department of Cognitive Science
Xiamen University

2 Intensity transformations

(2) Log Transformations

$$s = c\log(1+r), \quad \text{(c is constant)}$$

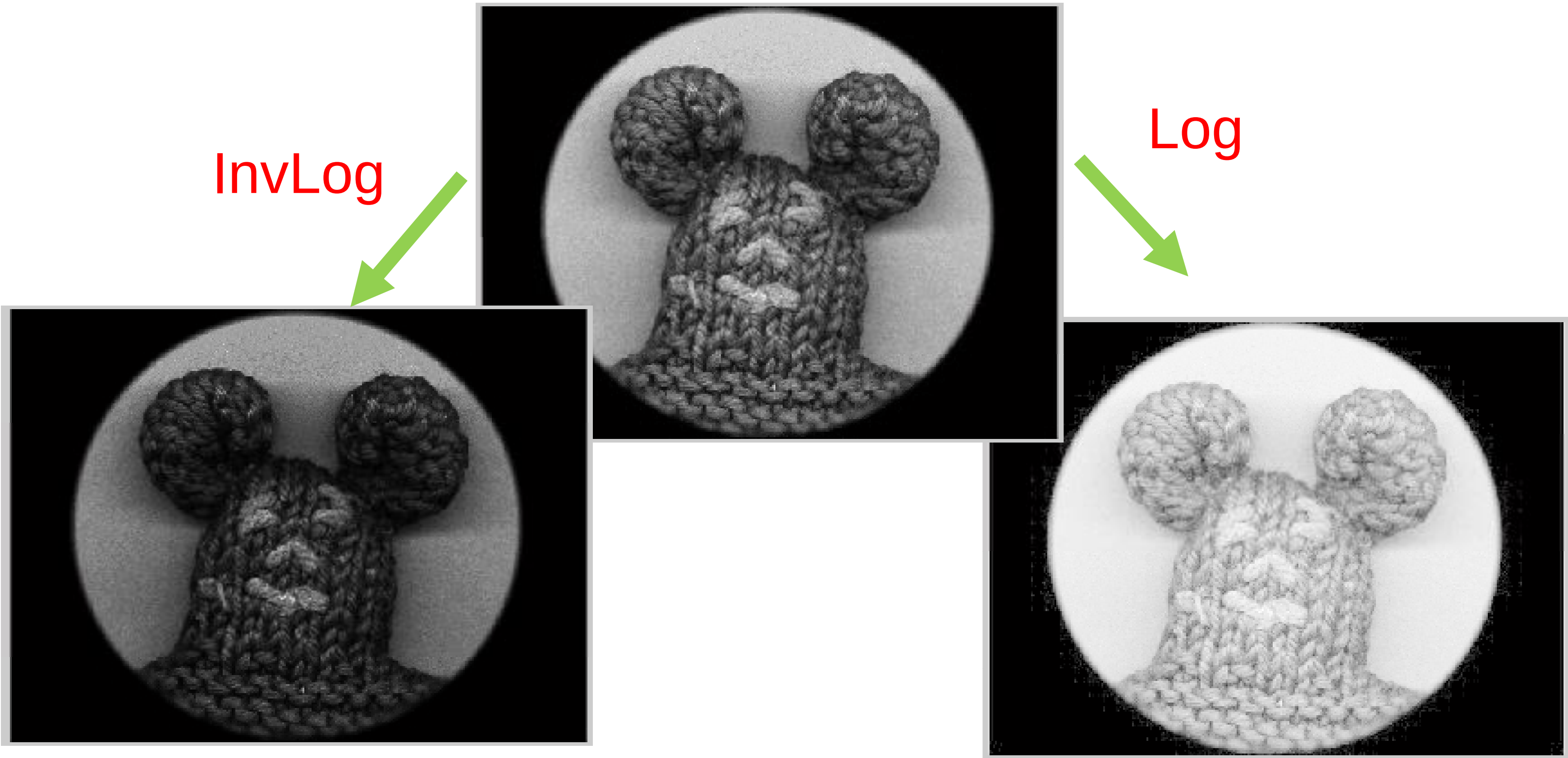


Map a narrow range of low intensity values in the input into a wider range of output levels.

Map a wide range of high intensity values in the input into a narrower range of output levels.

2 Intensity transformations

(2) Log Transformations



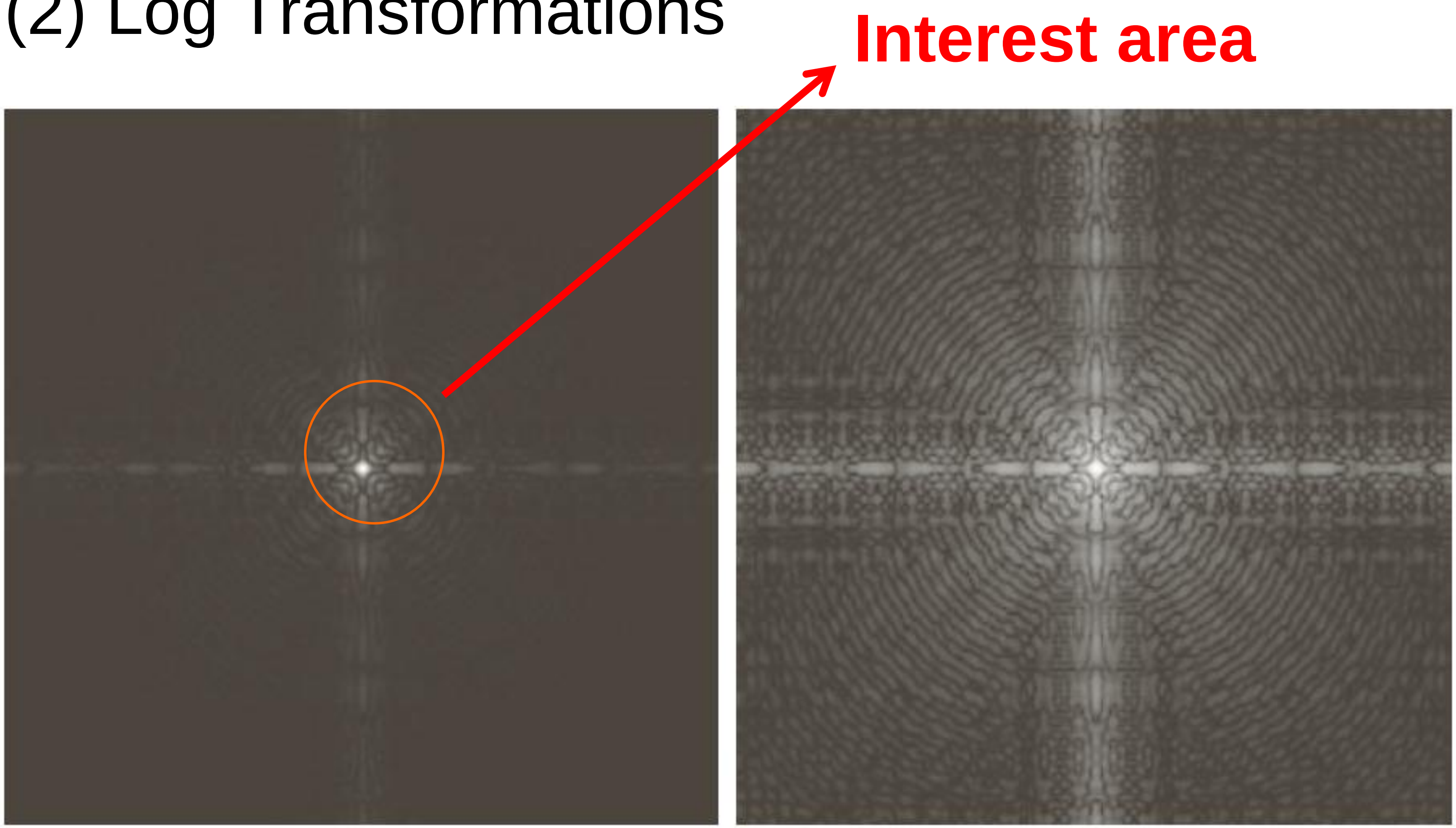
Suited for:

a. Expand values of dark pixels.
b. Compresses the dynamic range.

Department of Cognitive Science
Xiamen University

2 Intensity transformations

(2) Log Transformations



a b

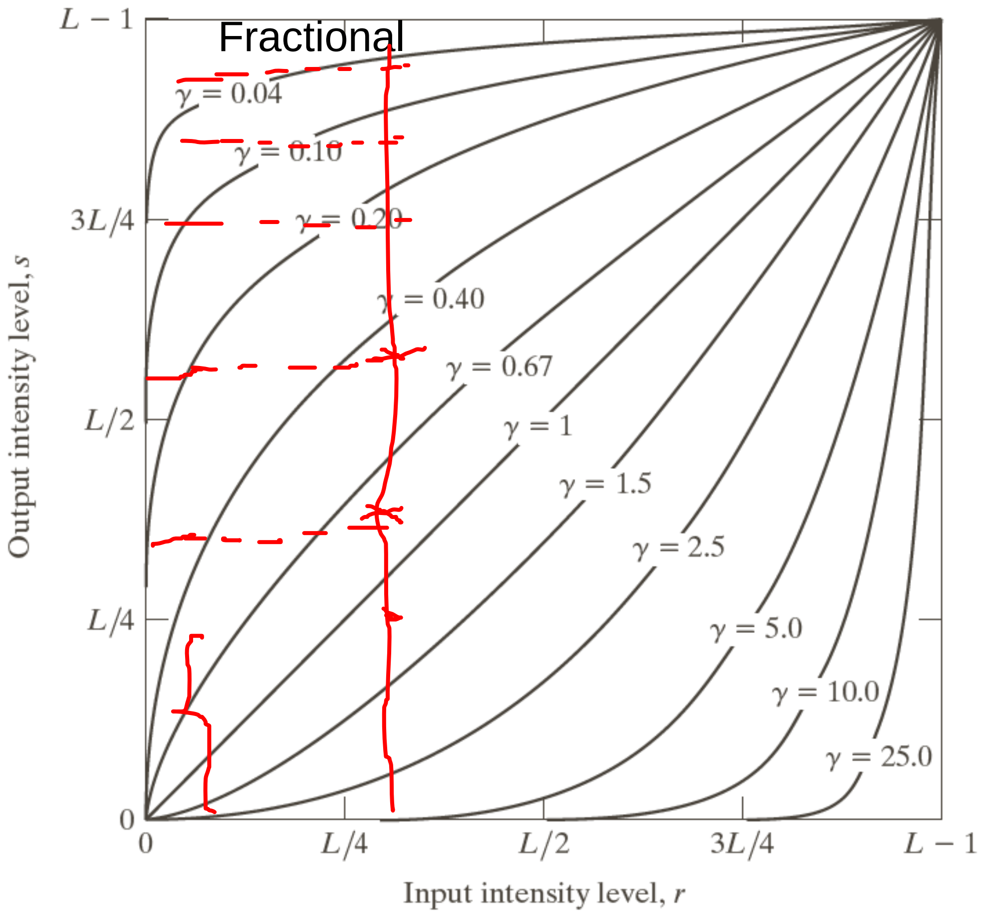
FIGURE 3.5 (a) Fourier spectrum. (b) Result of applying the log transformation in Eq. (3.2-2) with $c = 1$.

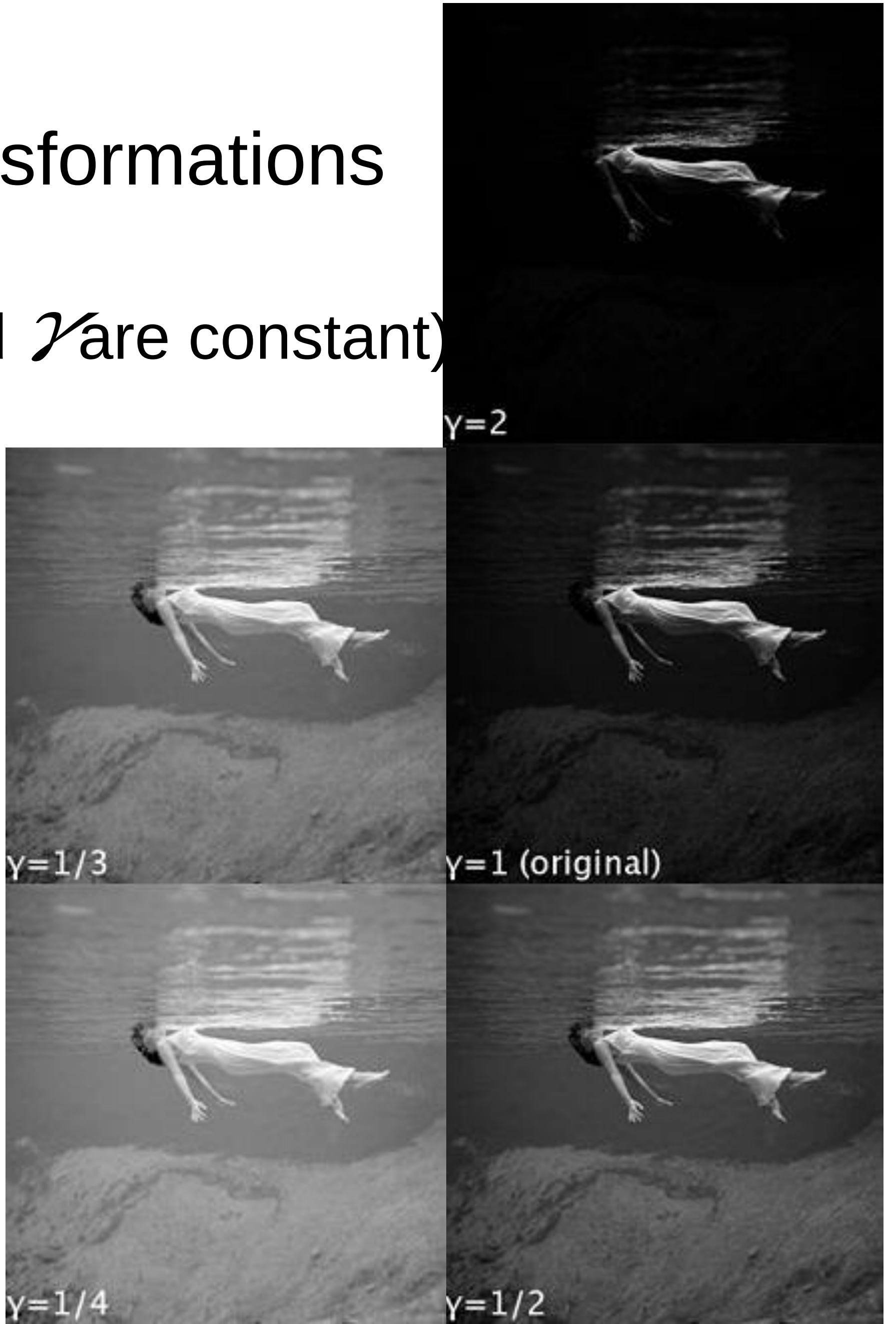
Fourier spectrum values range from 0 to 10^6 or higher. Image display systems can't reproduce such a wide range of intensity values. Scaling this new range linearly and displaying the spectrum in the 8-bit display.

2 Intensity transformations

(3) Power-law (Gamma) Transformations

$$s = cr^{\gamma}$$ (c and γ are constant)





2 Intensity transformations

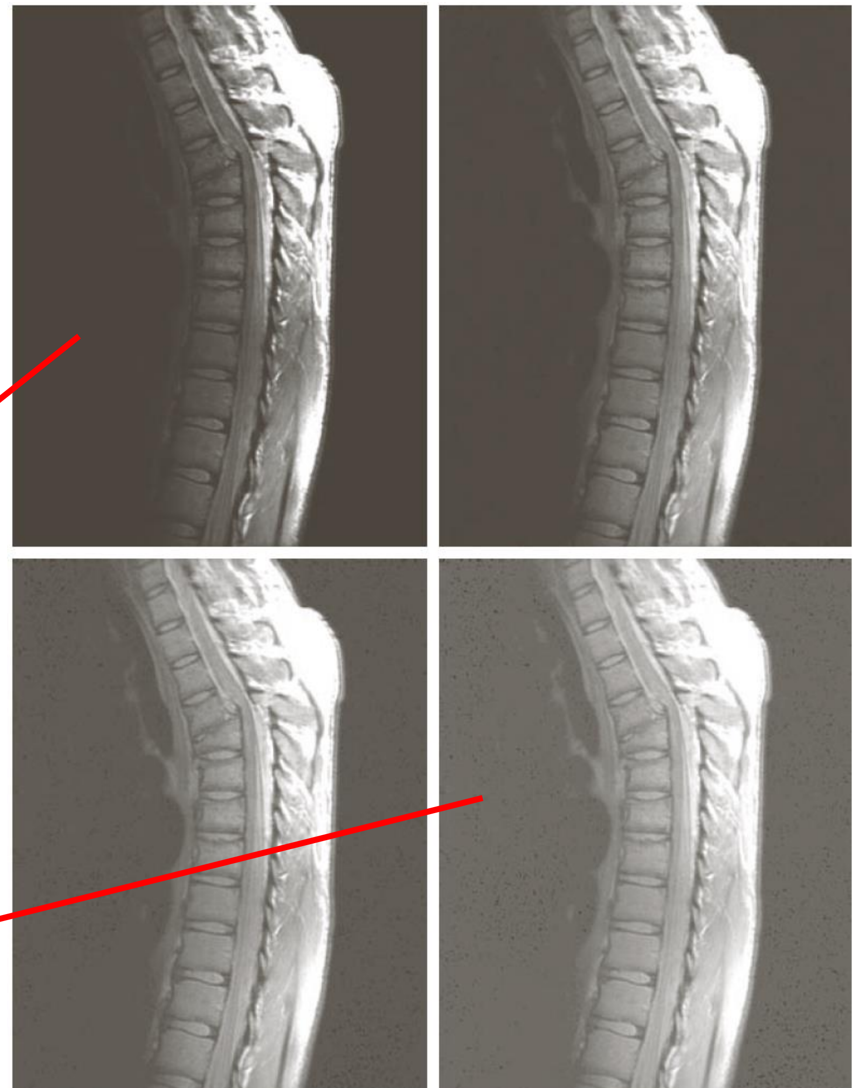
(3) Power-law (Gamma) Transformations

Contrast adjusting

Intensity levels
expansion needed

γ ?

Contrast decrease
Washed out!



a b
c d

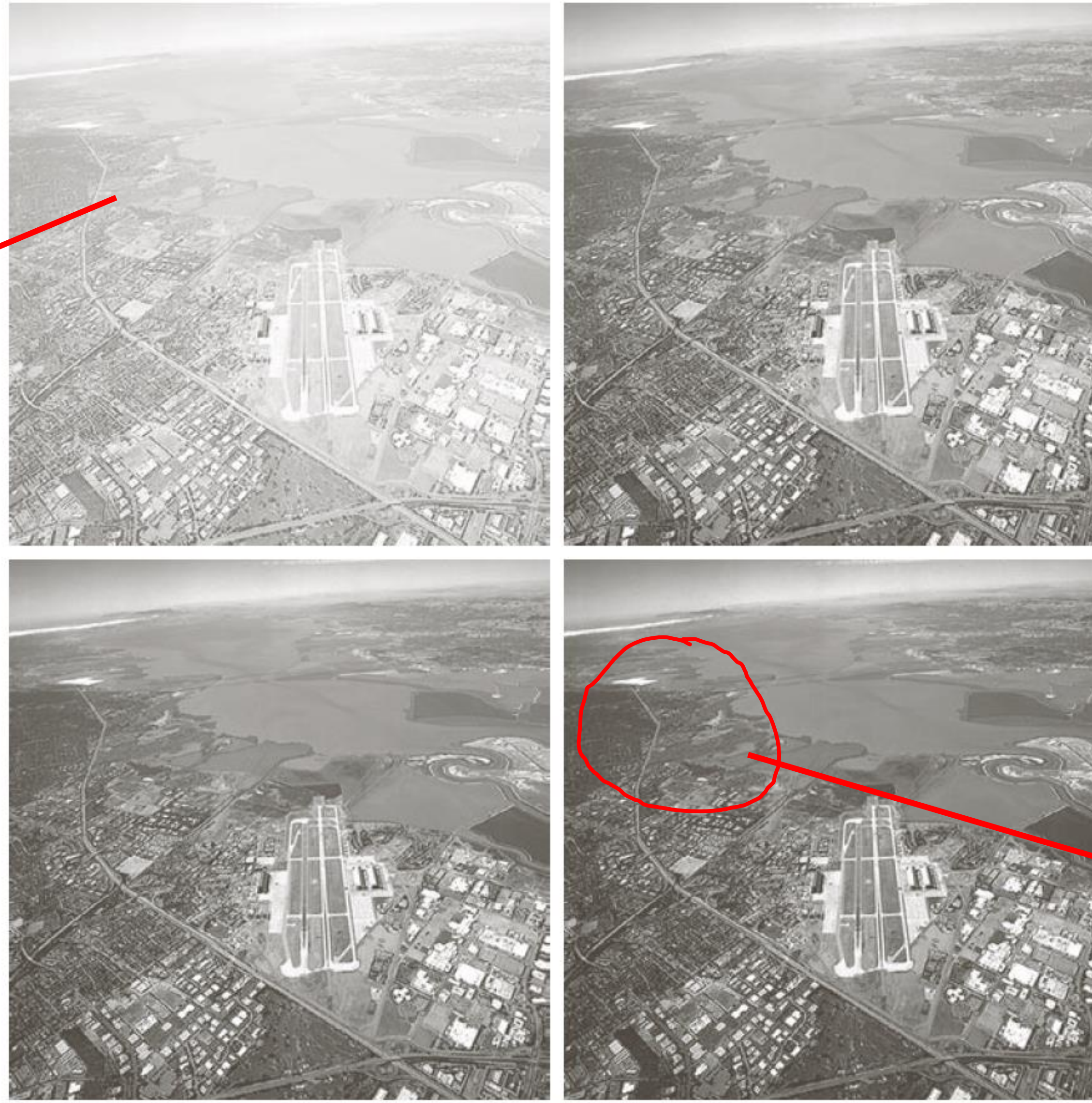
FIGURE 3.8 (a) Magnetic resonance image (MRI) of a fractured human spine. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3, respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Department of Cognitive Science
Xiamen University

Gamma Correction

Intensity levels compression needed

γ ?



a b
c d

FIGURE 3.9 (a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0,$ and 5.0, respectively. (Original image for this example courtesy of NASA.)

Detail is lost

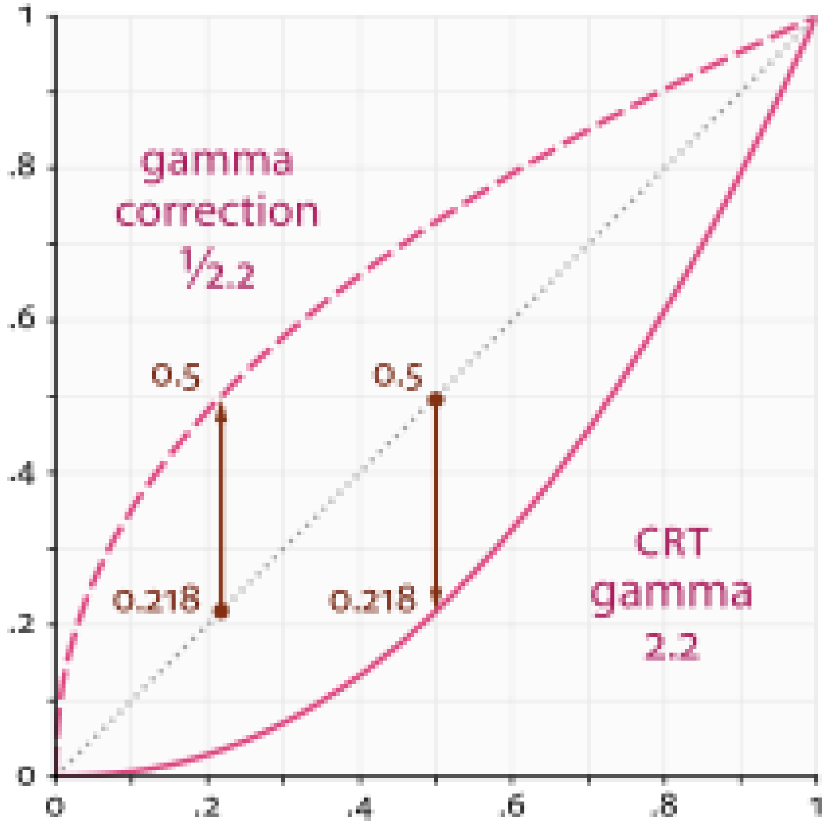
Why Gamma Correction?

Devices for image capture, printing and display respond according to a power law.

Why Gamma Correction?

— CRT monitor

Intensity-to-voltage response:
a power function with exponent varying from 1.8~2.5



Q: CRT output becomes darker or brighter?

Department of Cognitive Science
Xiamen University

Why Gamma Correction?

— CRT monitor

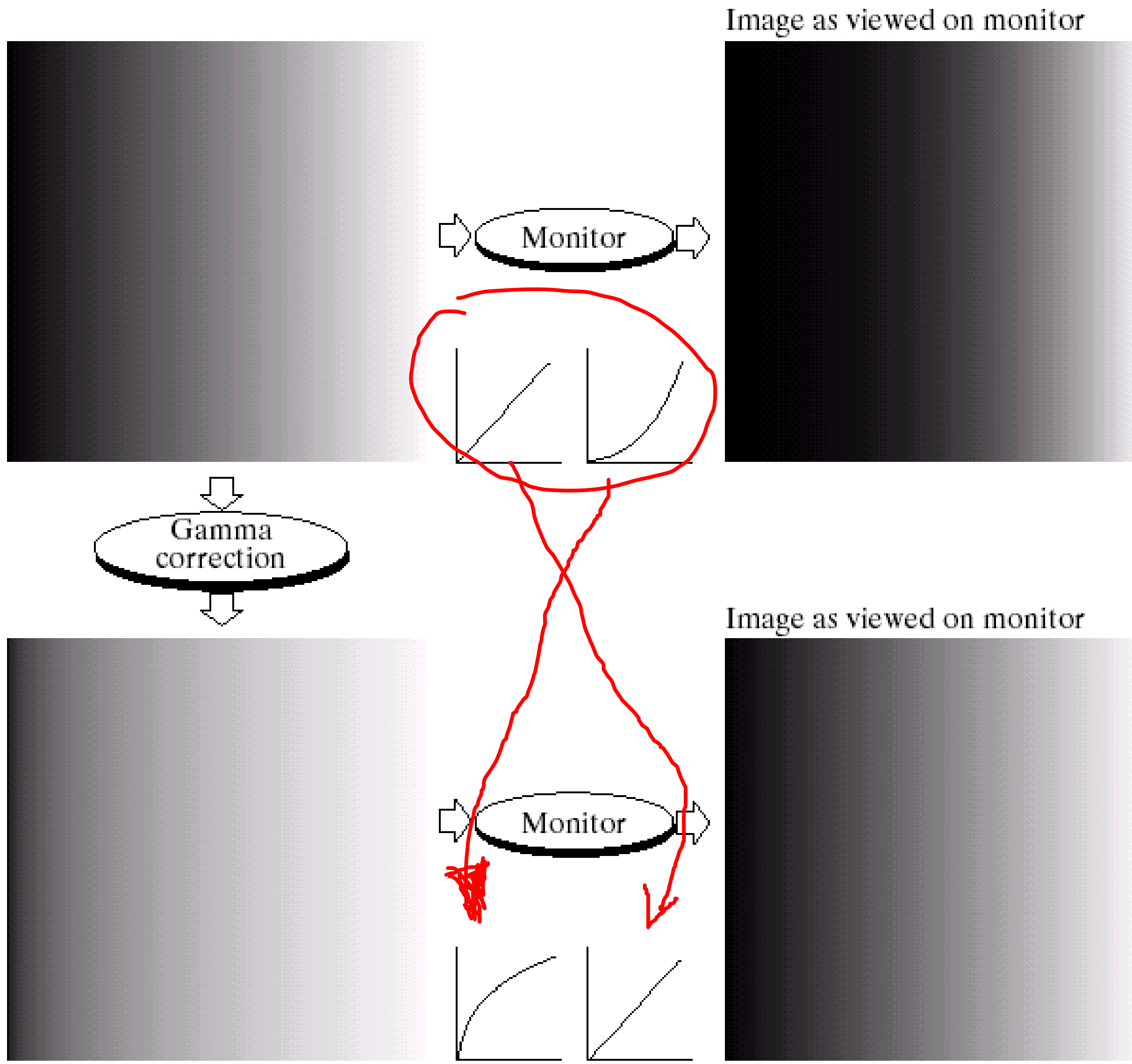


FIGURE 3.7 (a) Linear-wedge gray-scale image. (b) Response of monitor to linear wedge. (c) Gamma-corrected wedge. (d) Output of monitor.

2 Intensity transformations

(4) Piecewise-Linear Transformations

Complementary of several methods

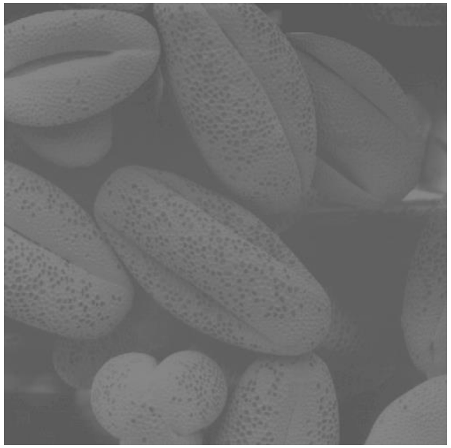
Contrast Stretching Transformation:

Expand the range of intensity levels to spans the full intensity range.

$$s = T(r) = mr + b$$

Low contrast image:

Poor illumination, lack of dynamic range in the image sensor and wrong setting of lens aperture.



Department of Cognitive Science
Xiamen University

2 Intensity transformations

(4) Piecewise-Linear Transformations

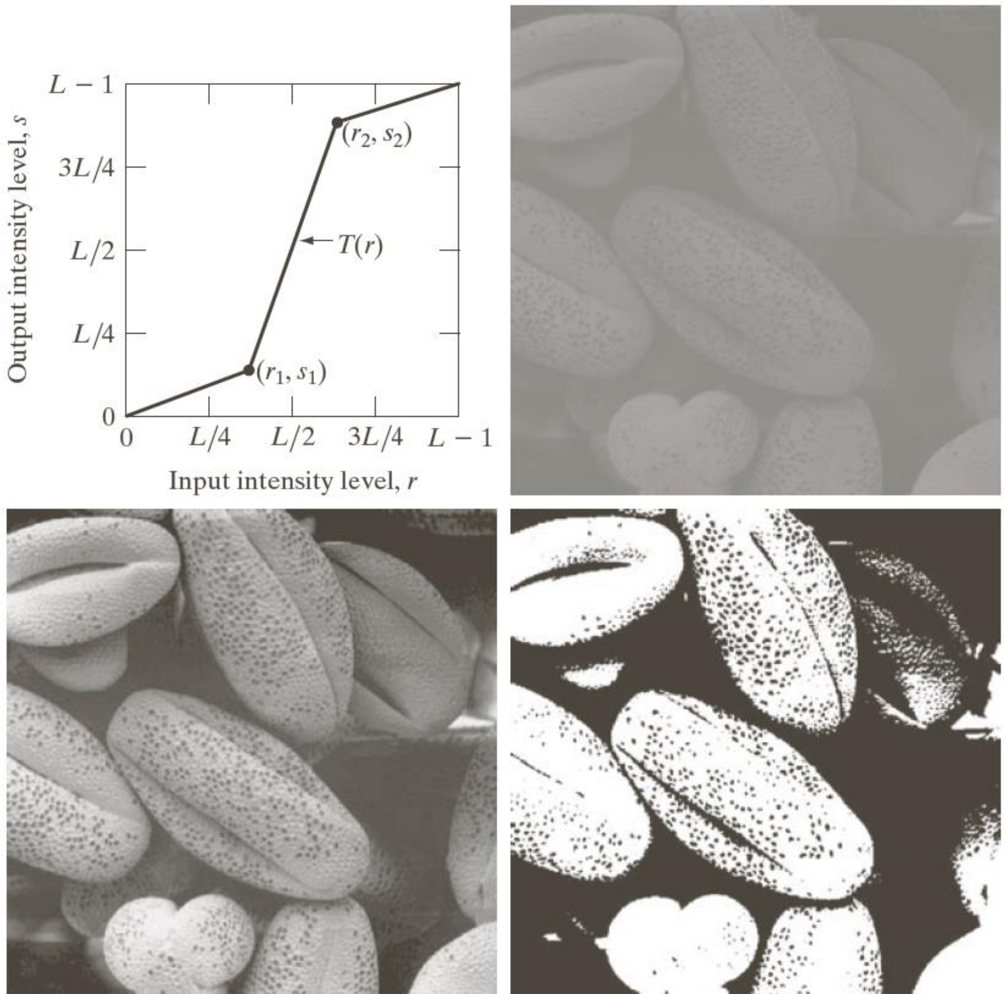
Contrast Stretching

$r_1 = s_1$, $r_2 = s_2$?

$r_1 = r_2$, $s_1 = 0$

$s_2 = L-1$?

Location of points controls the shape of transformation of function.

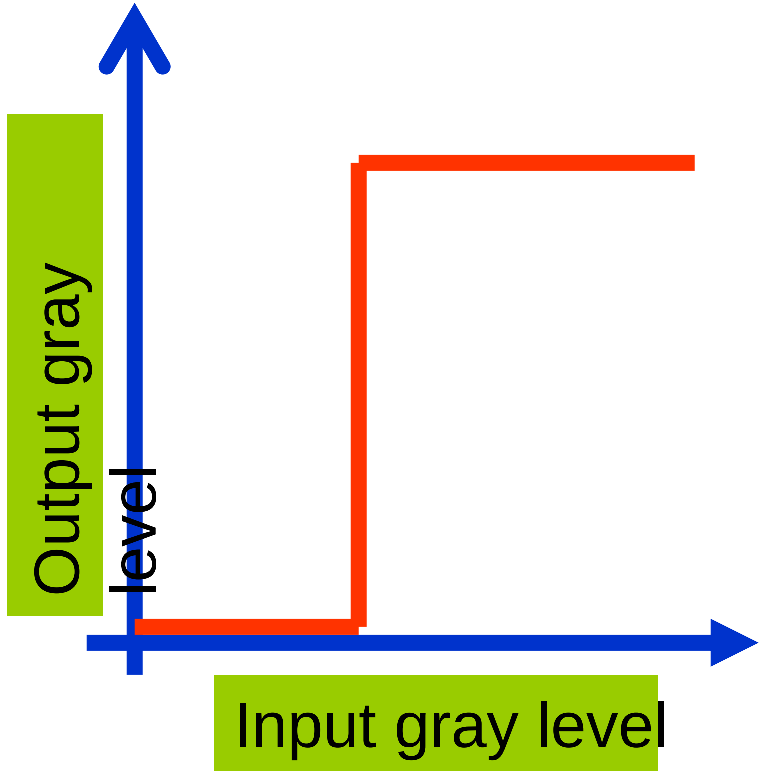


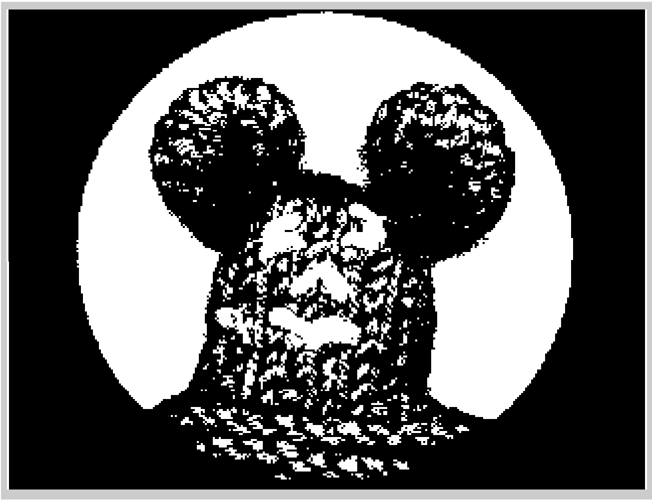
$$(r_1, s_1) = (r_{\min}, 0),$$
$$(r_2, s_2) = (r_{\max}, L-1)$$

$$(r_1, s_1) = (m, 0),$$
$$(r_2, s_2) = (m, L-1)$$

2 Intensity transformations

(4) Piecewise-Linear Transformations





Department of Cognitive Science
Xiamen University

2 Intensity transformations

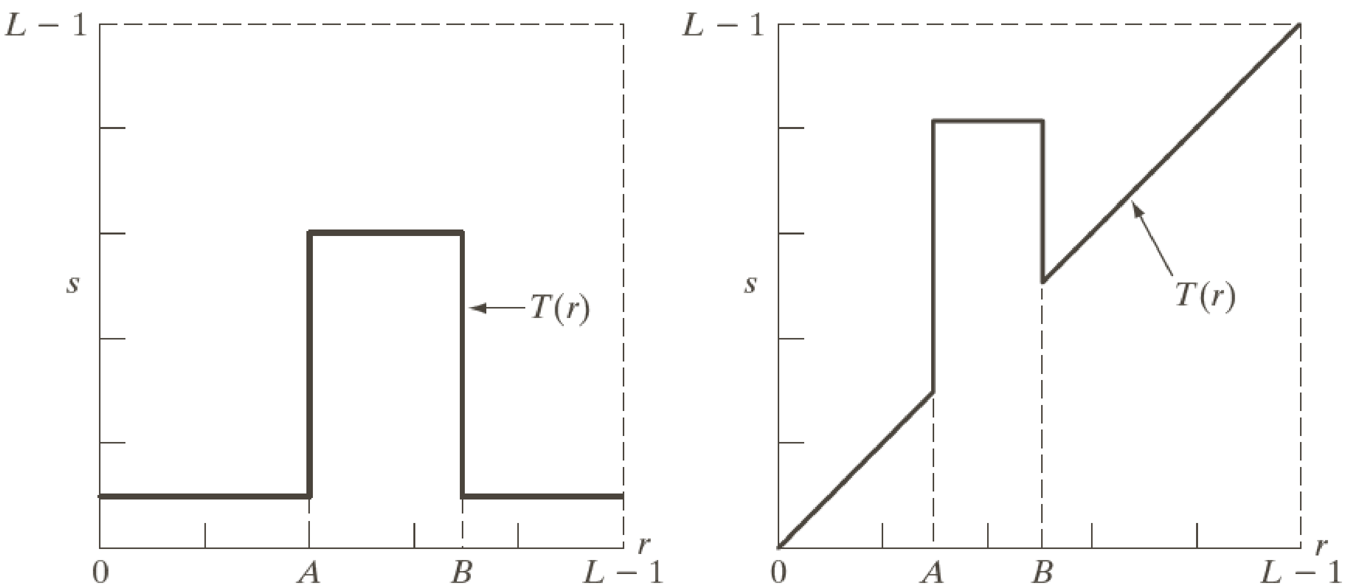
(4) Piecewise-Linear Transformations

Intensity-level slicing:

Enhancing certain features by: 1) one value all the values interested, and in another all other intensities. 2) brightens values interested, and leaves others unchanged.

a b

FIGURE 3.11 (a) This transformation highlights intensity range $[A, B]$ and reduces all other intensities to a lower level. (b) This transformation highlights range $[A, B]$ and preserves all other intensity levels.



2 Intensity transformations

(4) Piecewise-Linear Transformations

Intensity-level slicing:

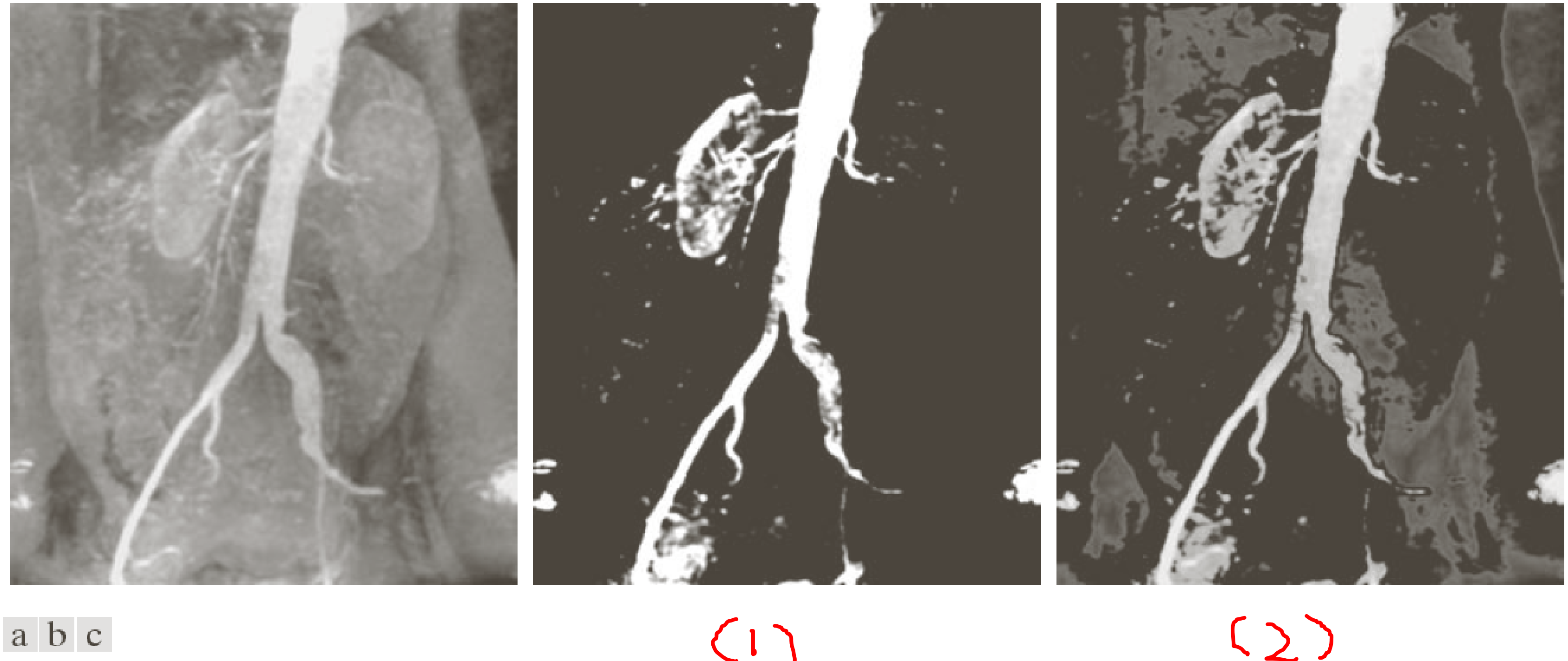


FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

3 Image Histograms

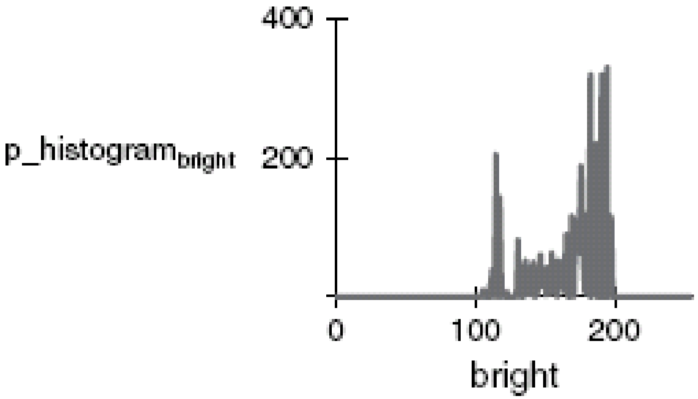
3.1 Definition

- A discrete function: the distribution of grey levels in the image
- Gray-level histogram: for each gray level, the number of pixels in the image have that gray level.

$$n_k = hist[r_k] = \sum_{f(x,y)=r_k} 1$$

- PDF: Normalized histogram (probability): $p_k = n_k / N$

(a) Image of eye



(b) Histogram of eye image

***x*-axis – values of intensities** ***y*-axis – their frequencies**

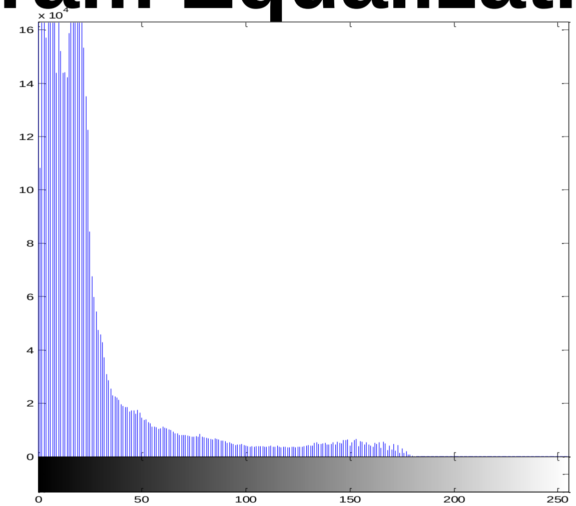
Histogram Equalization

Histogram equalization : is a method which increases the dynamic range of the gray-levels in a low-contrast image to cover full range of gray-levels.

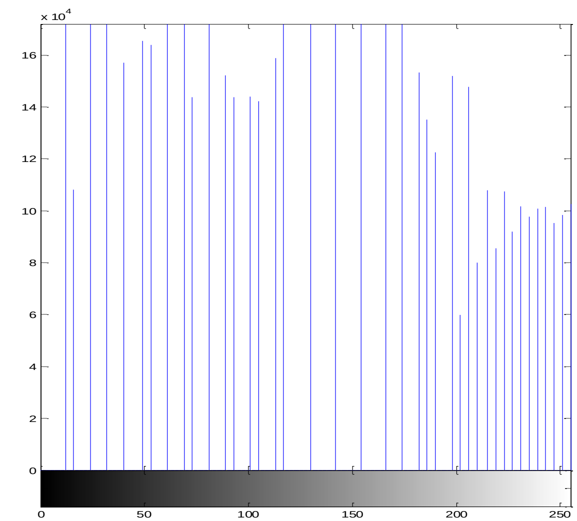
How-to-Do: is achieved by having a transformation function which is the Cumulative Distribution Function (CDF) of a given PDF of gray-levels in a given image.

Department of Cognitive Science
Xiamen University

Histogram Equalization

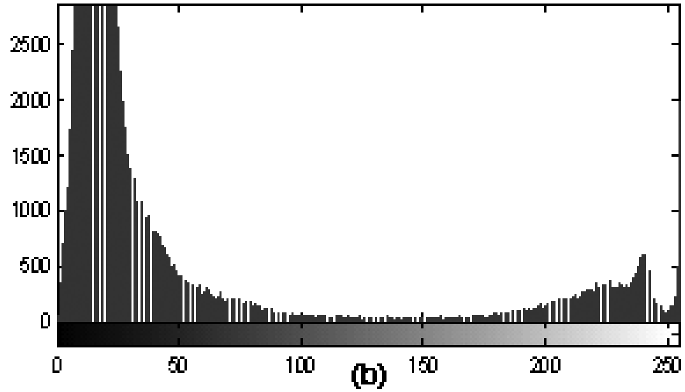


Histogram Equalization

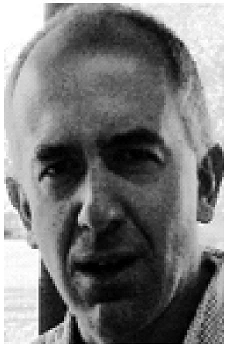


Histogram Equalization

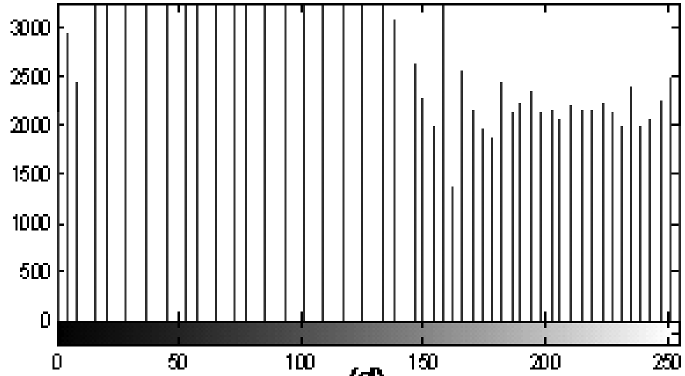
(a)



(b)

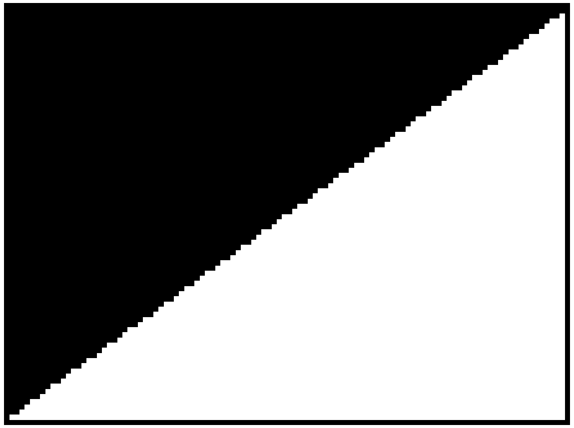


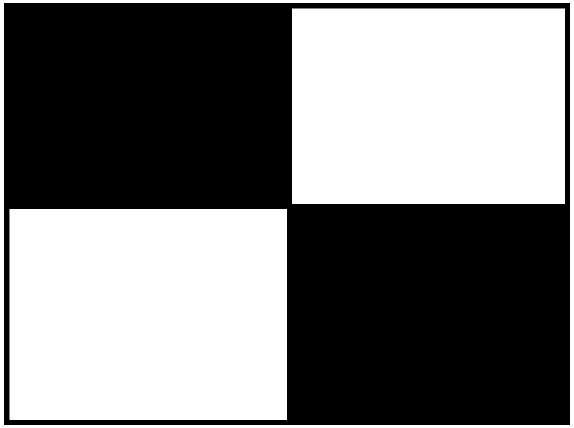
(c)



(d)

(a) A face image from the CALTECH face database, (b) its histogram, (c) the equalized face image using HE, (d) and its respective histogram.

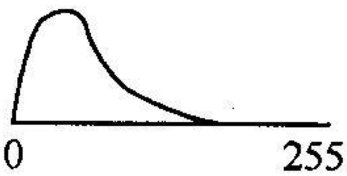




- Q:Same histogram?
- Does histogram show the pixel location information in the image?

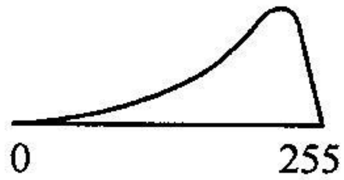
Histogram gives information about：

Bright or dark? Clear details? Dynamic range…



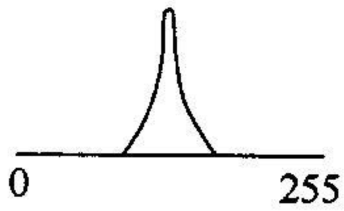
(a) 图象总体偏暗

Too dark



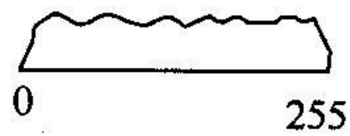
(b) 图象总体偏亮

Too bright



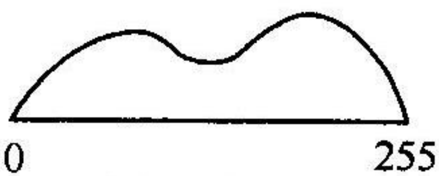
(c) 图象动态范围小，细节不够清楚

Bad details



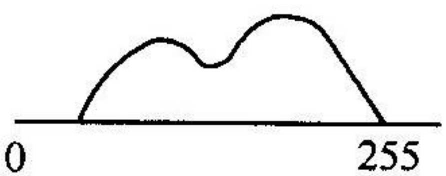
(d) 图象灰度分布均匀，清晰明快

Clear details



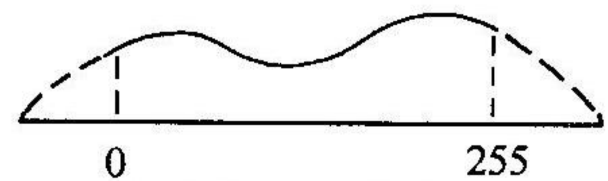
(e) 图象动态范围适中

Medium dynamic range



(f) 图象动态范围偏小

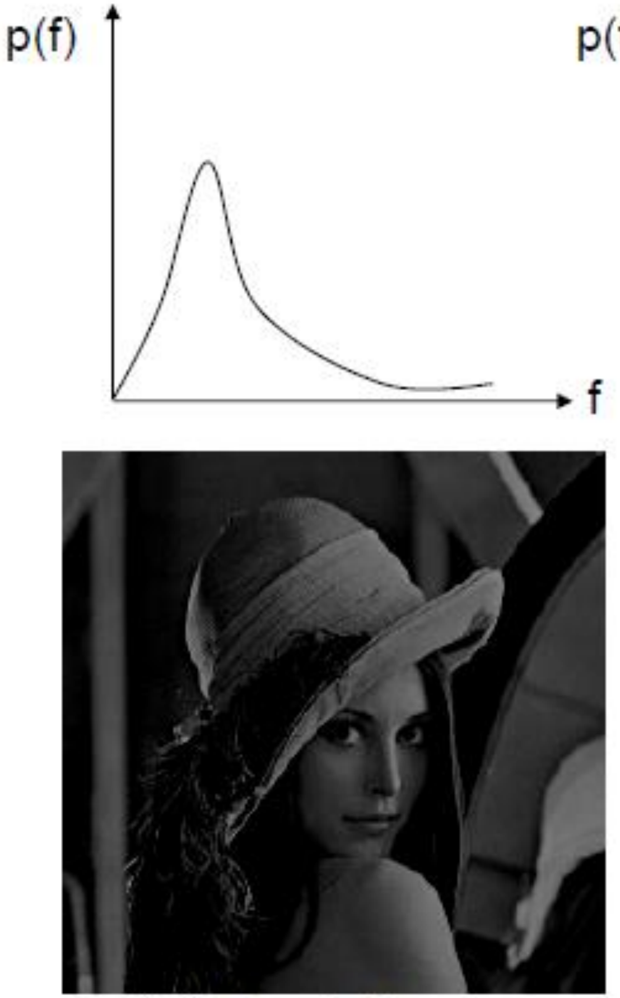
Small dynamic range



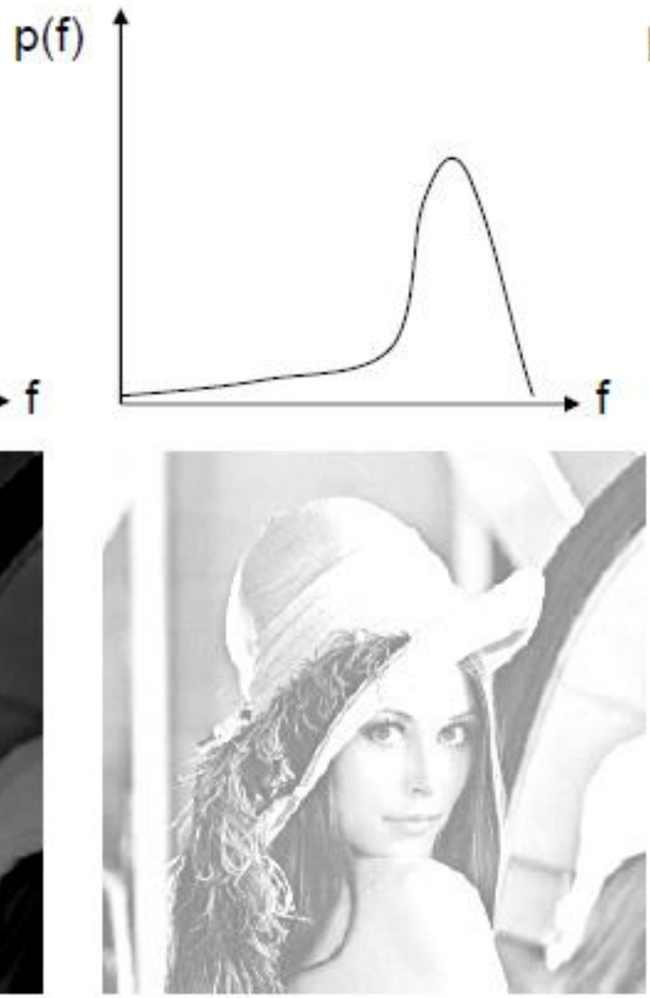
(g) 图象动态范围偏大

Wide dynamic range

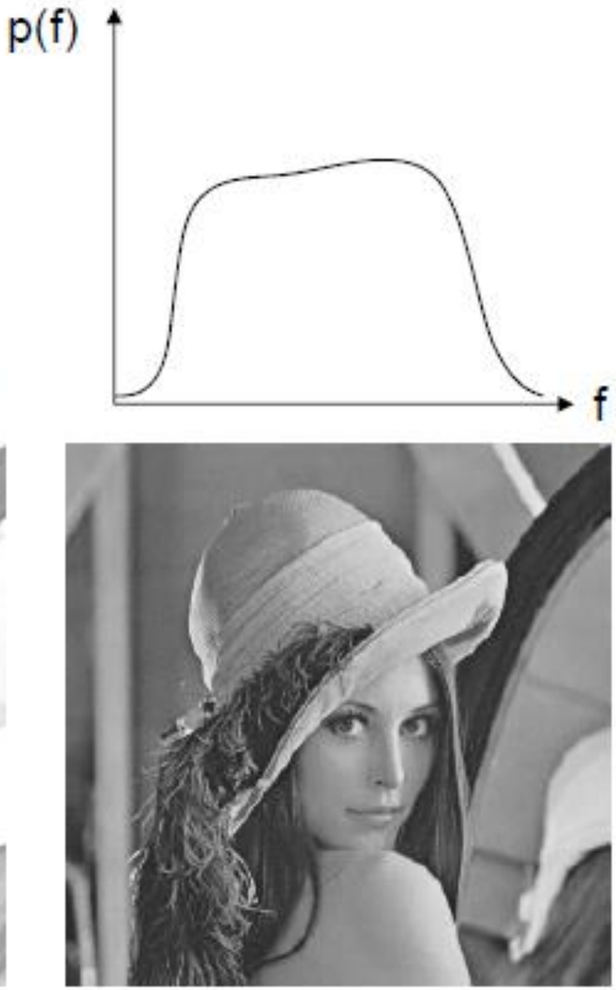
Examples of Histograms



(a) Too dark

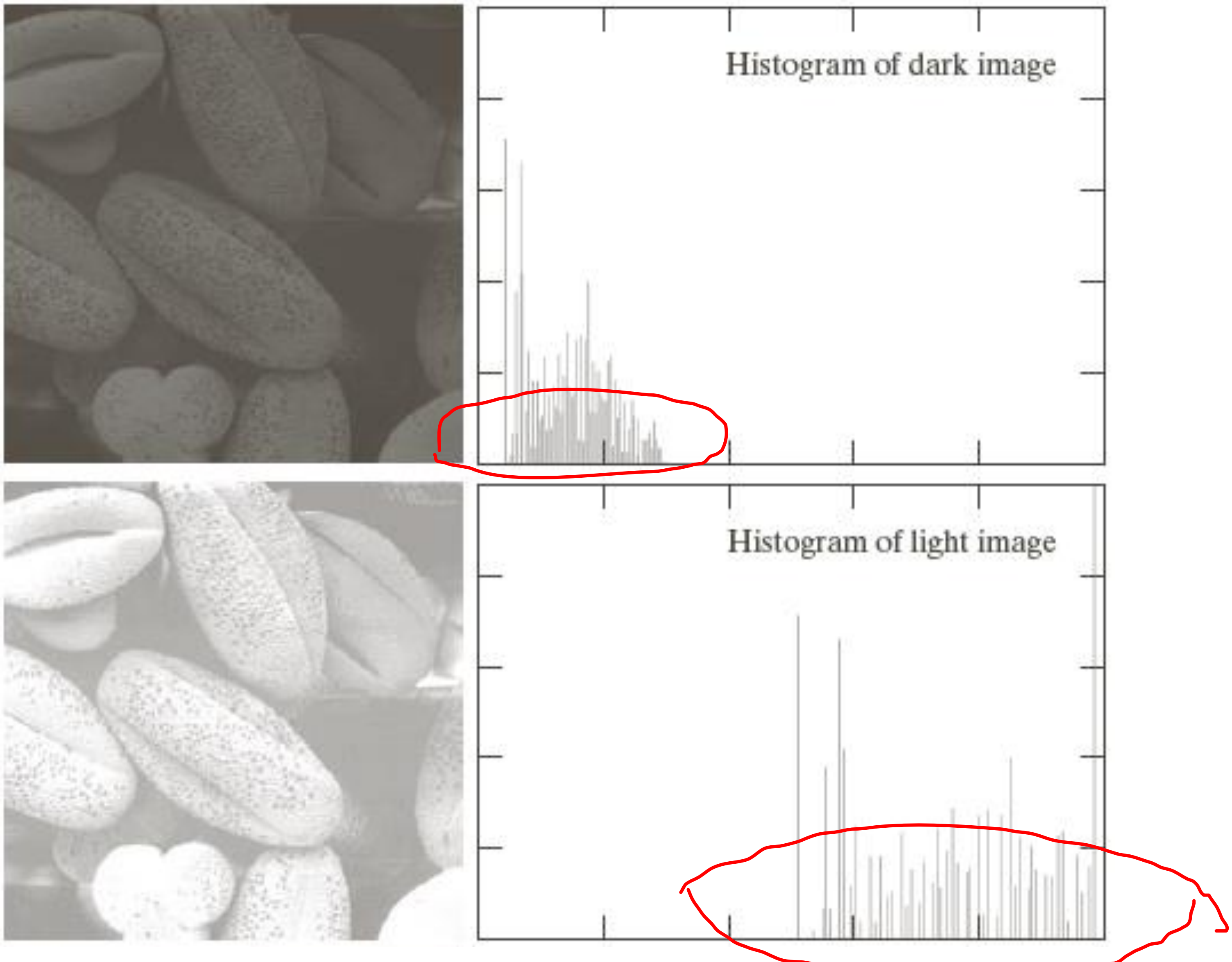


(b) Too bright

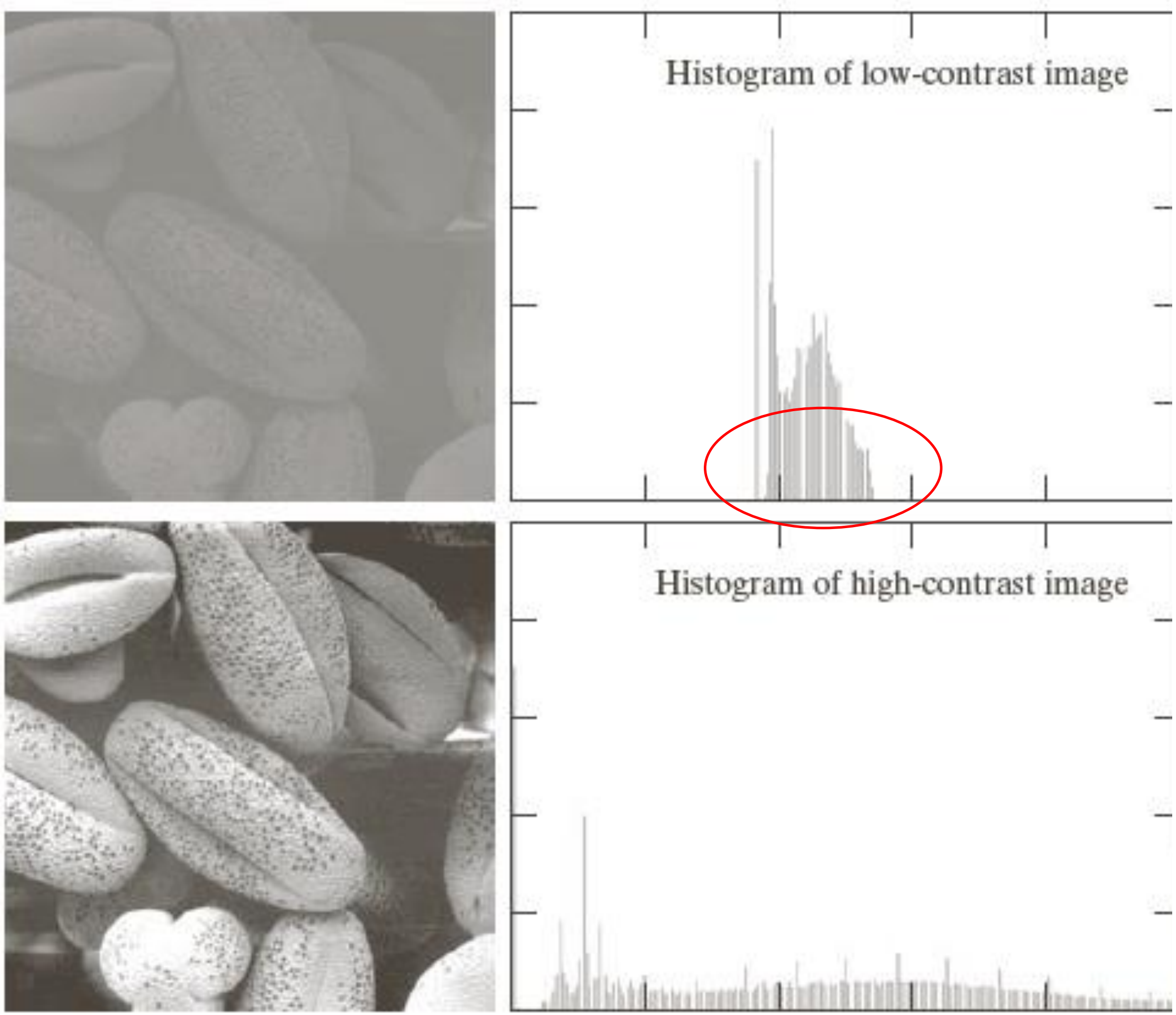


(c) Well balanced

histogram distribution



histogram distribution



Cover wide range of intensity

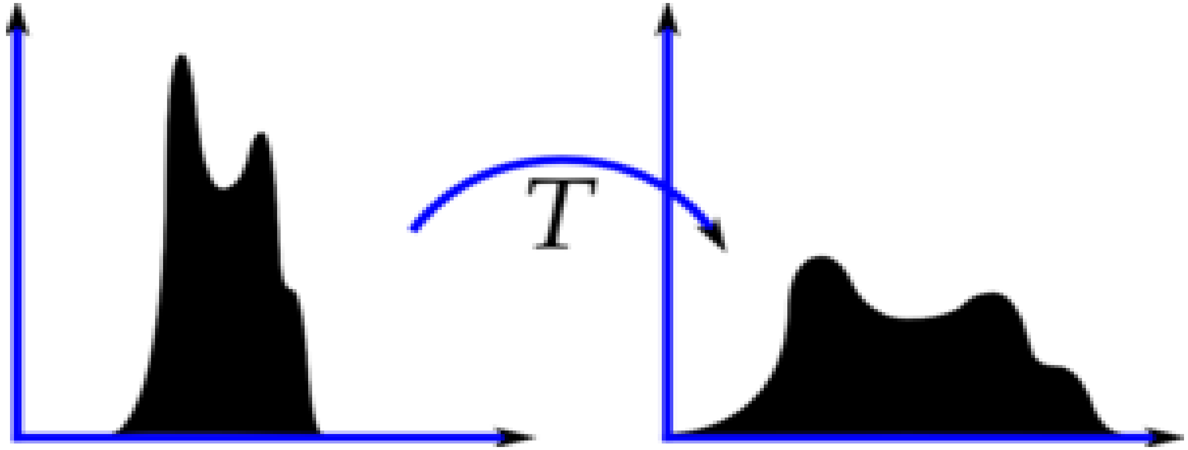
3.2 Histogram manipulation

What we want:

Find a gray level transformation function T → image f → Histogram of T(f) is 'equalized' distribution

How?

The image consists of an equal number of pixels for every gray-value, the histogram is constant !



3.2.1 Histogram Equalization

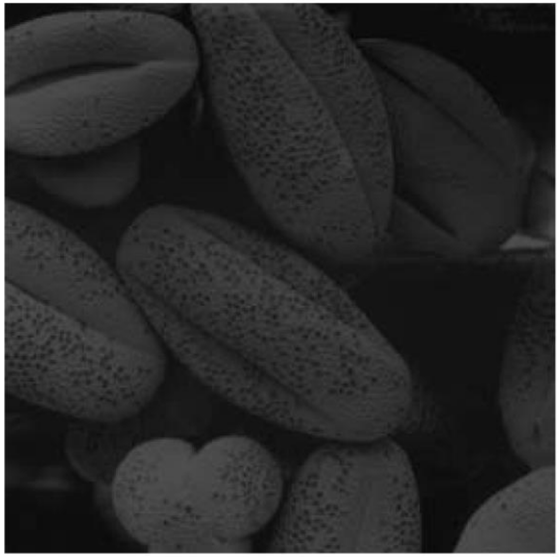
To make the *PDF* of the transformed image uniform,

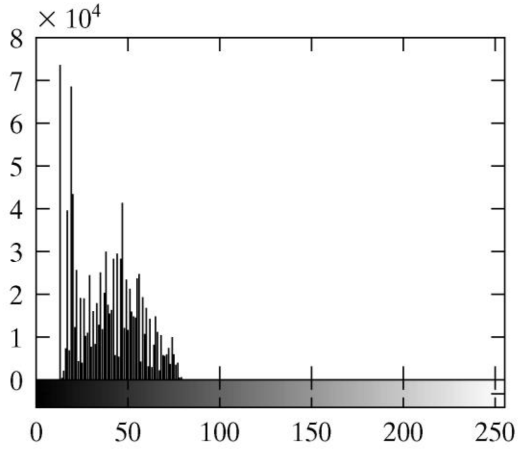
the formula for histogram equalization is given by

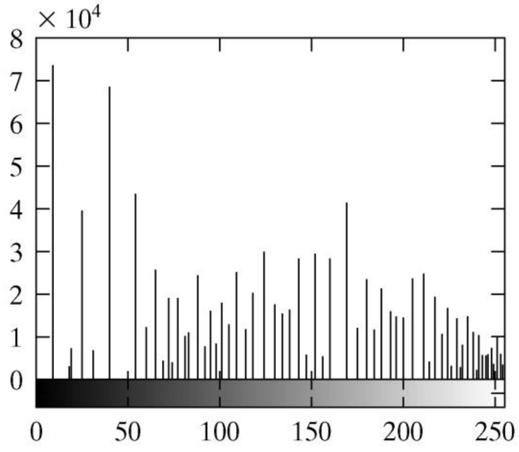
$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k}\frac{n_j}{n} = (L-1)\sum_{j=0}^{k} p_r(r_j)$$

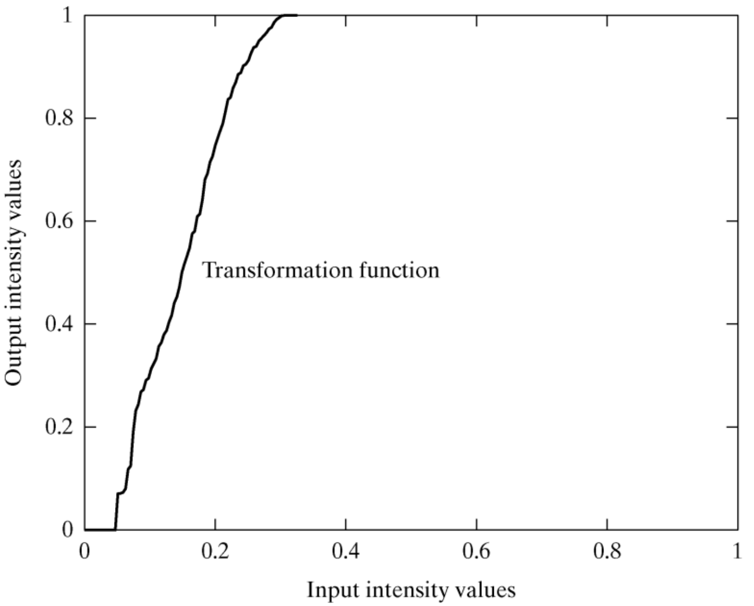
where

r_k: input intensity
s_k: processed intensity
k: the intensity range (e.g 0.0 – 1.0)
n_j: the frequency of intensity j
n: the sum of all frequencies
$p_r(r_j)$: the probability density function (*PDF*)
$T(r_k)$: the cumulative distribution function (CDF)




× 10^4
8
7
6
5
4
3
2
1
0
0
50
100
150
200
250

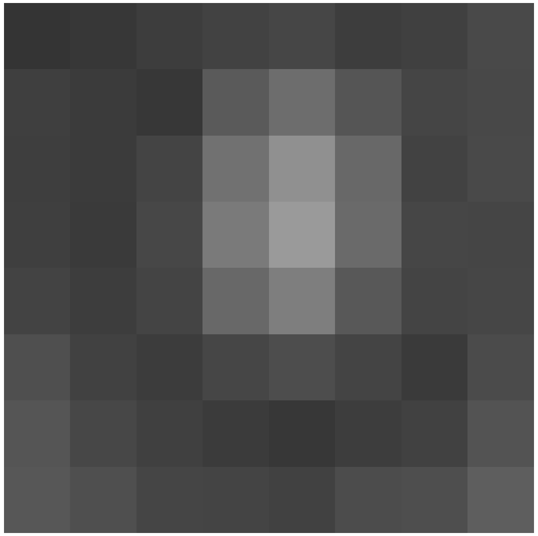

× 10^4
8
7
6
5
4
3
2
1
0
0
50
100
150
200
250


Output intensity values
Input intensity values
Transformation function
1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1

Histogram Equalization example 1(from wiki)

$$\begin{bmatrix} 52 & 55 & 61 & 66 & 70 & 61 & 64 & 73 \\ 63 & 59 & 55 & 90 & 109 & 85 & 69 & 72 \\ 62 & 59 & 68 & 113 & 144 & 104 & 66 & 73 \\ 63 & 58 & 71 & 122 & 154 & 106 & 70 & 69 \\ 67 & 61 & 68 & 104 & 126 & 88 & 68 & 70 \\ 79 & 65 & 60 & 70 & 77 & 68 & 58 & 75 \\ 85 & 71 & 64 & 59 & 55 & 61 & 65 & 83 \\ 87 & 79 & 69 & 68 & 65 & 76 & 78 & 94 \end{bmatrix}$$

Original Image



Histogram Equalization Example 1(from wiki)

Histogram (PDF)

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52	1	64	2	72	1	85	2	113	1
55	3	65	3	73	2	87	1	122	1
58	2	66	2	75	1	88	1	126	1
59	3	67	1	76	1	90	1	144	1
60	1	68	5	77	1	94	1	154	1
61	4	69	3	78	1	104	2		
62	1	70	4	79	2	106	1		
63	2	71	2	83	1	109	1		

Histogram Equalization Example 1(from wiki)

cumulative distribution function (CDF)

Value	cdf	Value	cdf	Value	cdf	Value	cdf	Value	cdf
52	1	64	19	72	40	85	51	113	60
55	4	65	22	73	42	87	52	122	61
58	6	66	24	75	43	88	53	126	62
59	9	67	25	76	44	90	54	144	63
60	10	68	30	77	45	94	55	154	64
61	14	69	33	**78**	**46**	104	57		
62	15	70	37	79	48	106	58		
63	17	71	39	83	49	109	59		

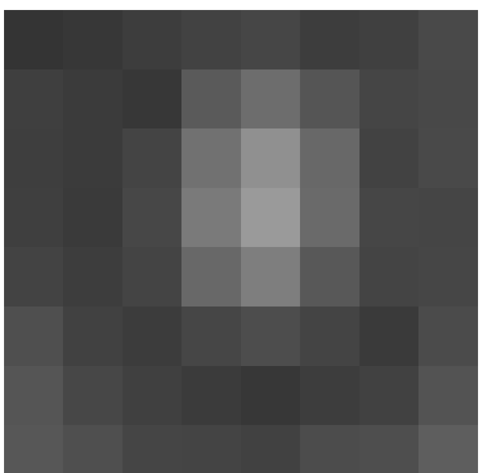
CDF shows the minimum value in the sub-image is 52 and the maximum value is 154. The CDF of 64 for value 154 coincides with the number of pixels in the image.

$$h(v) = \text{round}\left(\frac{cdf(v) - cdf_{min}}{(M \times N) - cdf_{min}} \times (L - 1)\right)$$

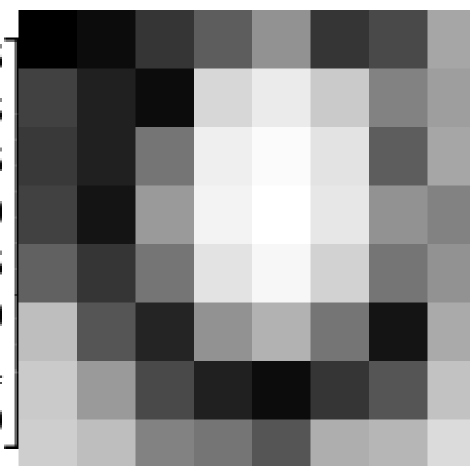
$$h(v) = \text{round}\left(\frac{cdf(v) - 1}{63} \times 255\right)$$

For example, the cdf of 78 is 46,the normalized value becomes:

$$h(78) = \text{round}\left(\frac{46 - 1}{63} \times 255\right) = \text{round}\left(0.714286 \times 255\right) = 182$$



$$\begin{bmatrix} 0 & 12 & 53 & 93 & 146 & 53 & 73 & 166 \\ 65 & 32 & 12 & 215 & 235 & 202 & 130 & 158 \\ 57 & 32 & 117 & 239 & 251 & 227 & 93 & 166 \\ 65 & 20 & 154 & 243 & 255 & 231 & 146 & 130 \\ 97 & 53 & 117 & 227 & 247 & 210 & 117 & 146 \\ 190 & 85 & 36 & 146 & 178 & 117 & 20 & 170 \\ 202 & 154 & 73 & 32 & 12 & 53 & 85 & 194 \\ 206 & 190 & 130 & 117 & 85 & 174 & 182 & 219 \end{bmatrix}$$



Histogram Equalization Example 2

200	50	100
50	100	50

Input	number	pdf	cdf	Output
50	3	3/6	3/6=**50%**	255*50%=**128**
100	2	2/6	(3+2)/6=**83.33%**	255*83.33%=**212**
200	1	1/6	(3+2+1)/6=**100%**	255*100%=**255**

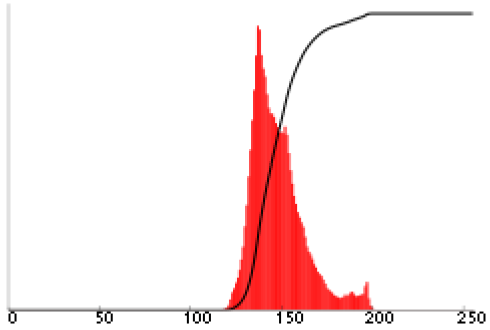
255	128	212
128	212	128

Department of Cognitive Science
Xiamen University

Histogram Equalization Example 3

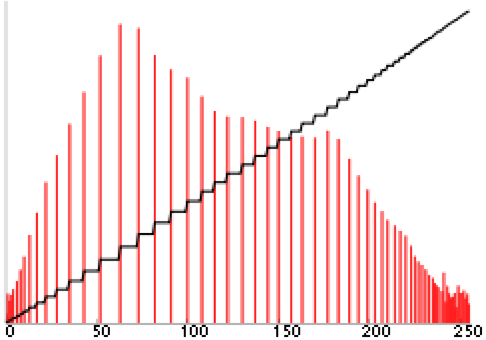
An un-equalized image

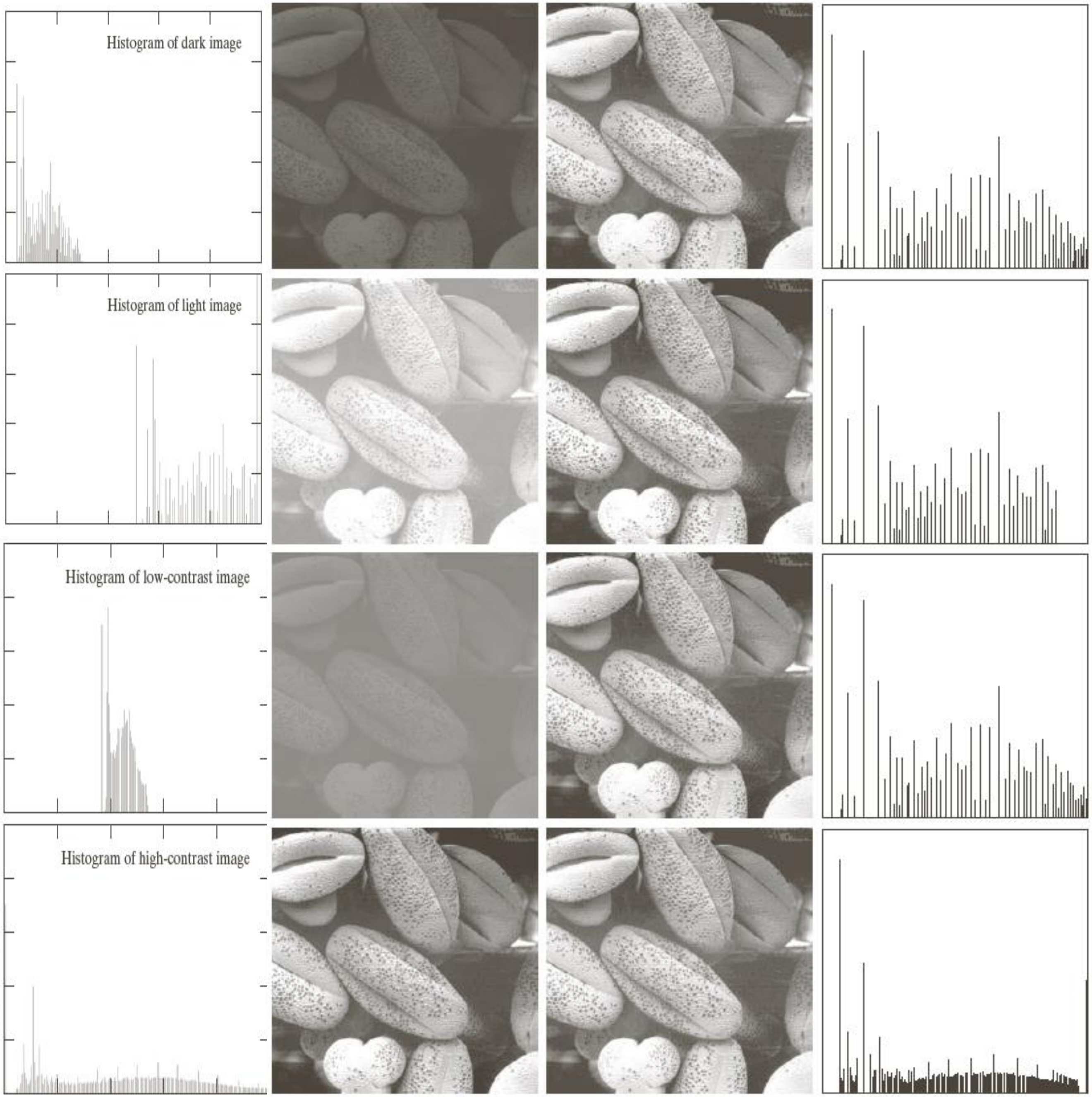
Corresponding histogram (red) and cumulative histogram (black)

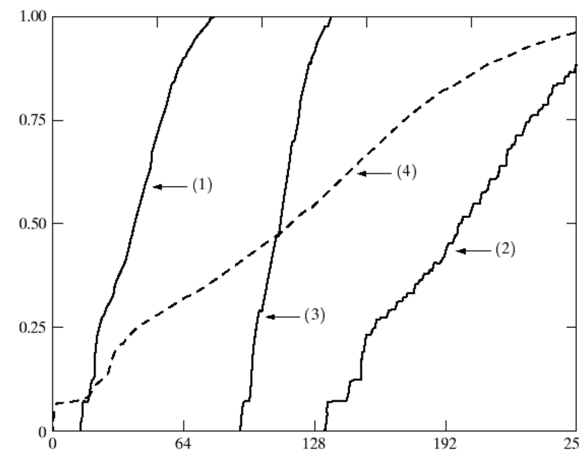


The same image after histogram equalization

Corresponding histogram (red) and cumulative histogram (black)




Histogram of dark image
Histogram of light image
Histogram of low-contrast image
Histogram of high-contrast image


1.00
0.75
0.50
0.25
0
0
64
128
192
(1)
(2)
(3)
(4)

3.2.2 Histogram Matching

— generate a processed image that has a specified histogram

Procedure:

1.Obtain $p_r(r_j)$ from the input image and then obtain the values of s_k, round the value to the integer range [0, L-1].

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

2.Use the specified PDF and obtain the transformation function $G(z_q)$, round the value to the integer range [0, L-1].

$$G(z_q) = (L-1)\sum_{i=0}^{q} p_z(z_i) = s_k$$

3.Mapping from s_k to z_q

$$z_q = G^{-1}(s_k)$$

Histogram Matching Example 1

Suppose that a 3-bit image (L=8) of size 64 × 64 pixels (MN = 4096) has the intensity distribution shown in the following table (on the left). Get the histogram transformation function and make the output image with the specified histogram, listed in the table on the right.

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

z_q	Specified $p_z(z_q)$	Actual $p_z(z_k)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11

Histogram Matching Example 1

Obtain the scaled histogram-equalized values,

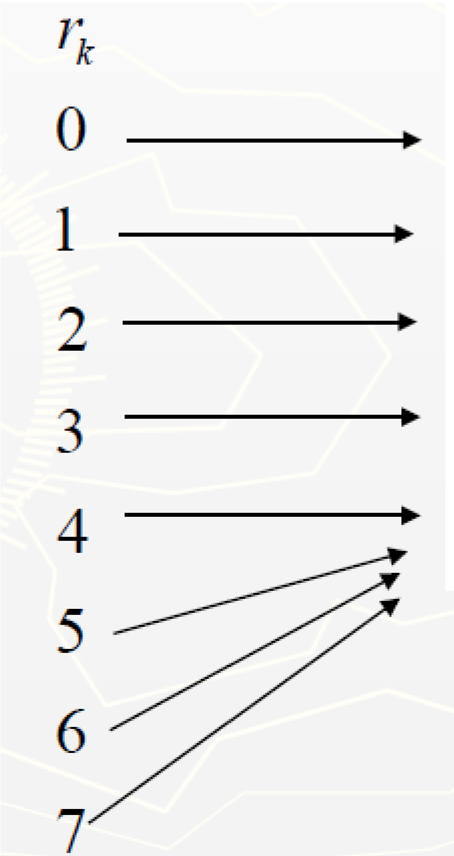
$$s_0 = 1, \quad s_1 = 3, \quad s_2 = 5, \quad s_3 = 6, \quad s_4 = 7,$$
$$s_5 = 7, \quad s_6 = 7, \quad s_7 = 7.$$

Compute all the values of the transformation function G,

$$G(z_0) = 7\sum_{j=0}^{0} p_z(z_j) = 0.00 \rightarrow 0$$

$$G(z_1) = 0.00 \rightarrow 0 \qquad G(z_2) = 0.00 \rightarrow 0$$
$$G(z_3) = 1.05 \rightarrow 1 \quad s_0 \qquad G(z_4) = 2.45 \rightarrow 2 \quad s_1$$
$$G(z_5) = 4.55 \rightarrow 5 \quad s_2 \qquad G(z_6) = 5.95 \rightarrow 6 \quad s_3$$
$$G(z_7) = 7.00 \rightarrow 7 \quad s_4 \; s_5 \; s_6 \; s_7$$

Histogram Matching Example 1



s_k	$\rightarrow$	z_q
1	$\rightarrow$	3
3	$\rightarrow$	4
5	$\rightarrow$	5
6	$\rightarrow$	6
7	$\rightarrow$	7

$r_k \rightarrow z_q$

$0 \rightarrow 3$

$1 \rightarrow 4$

$2 \rightarrow 5$

$3 \rightarrow 6$

$4 \rightarrow 7$

$5 \rightarrow 7$

$6 \rightarrow 7$

$7 \rightarrow 7$

Department of Cognitive Science
Xiamen University

Histogram Matching Example 1

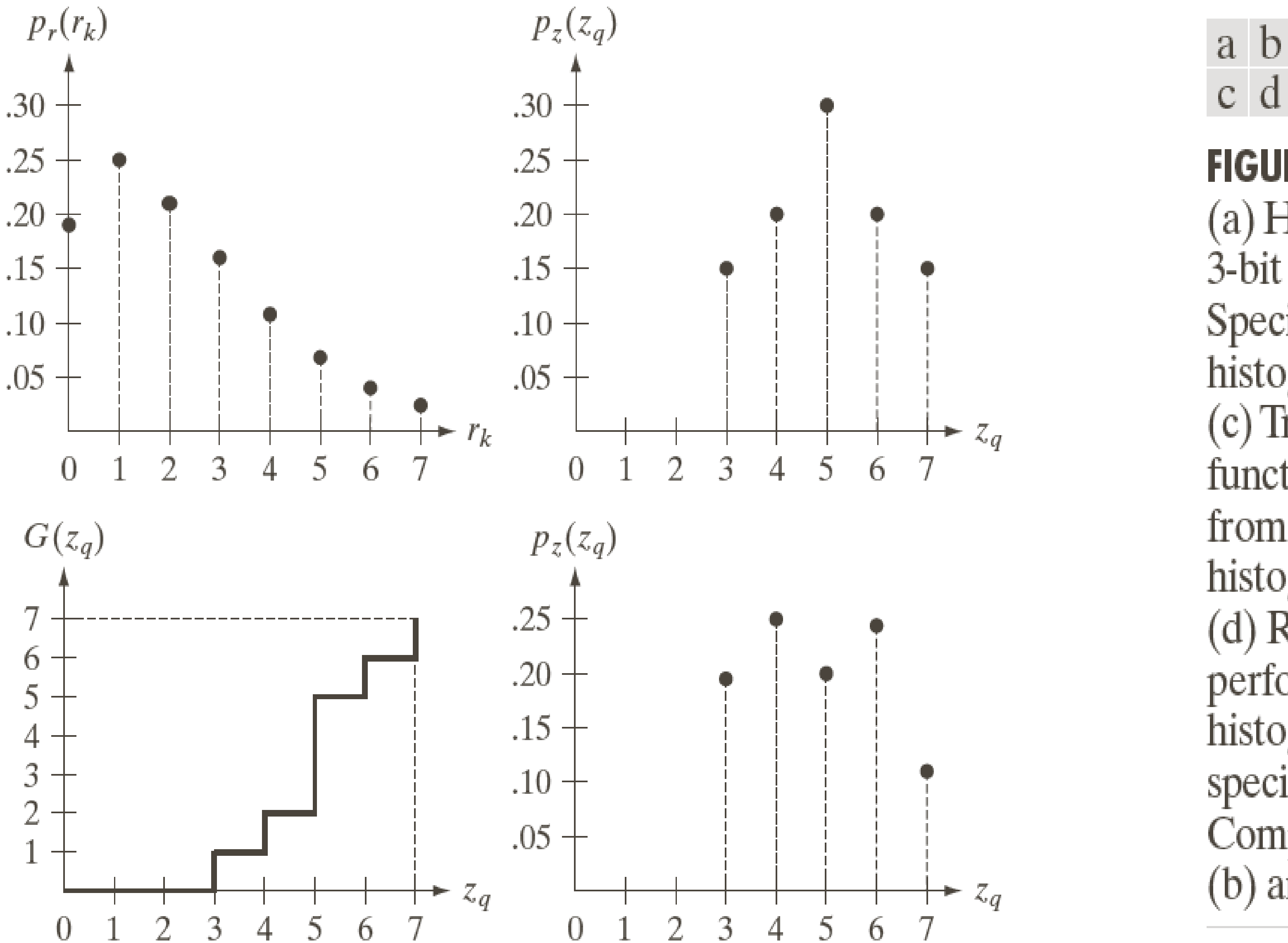
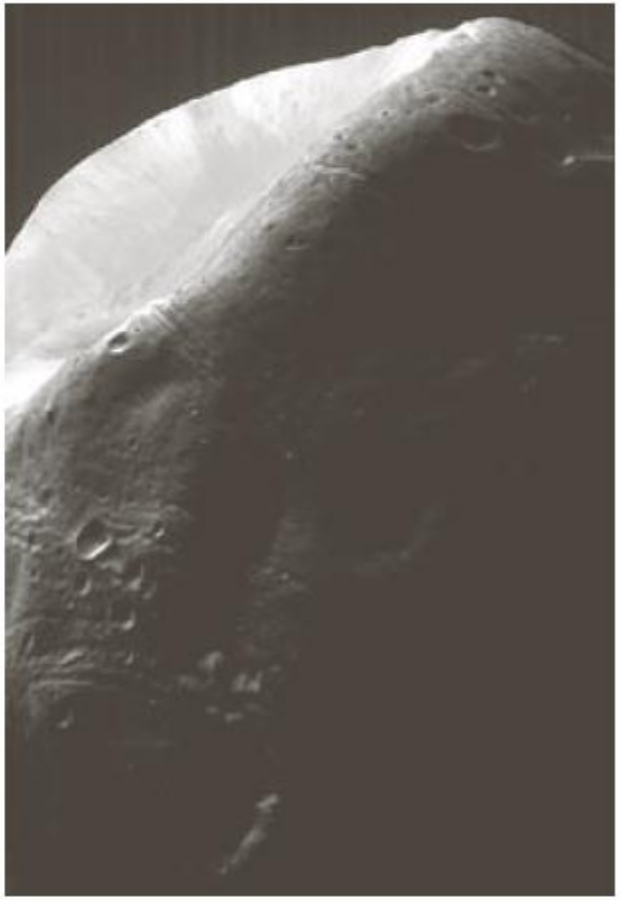
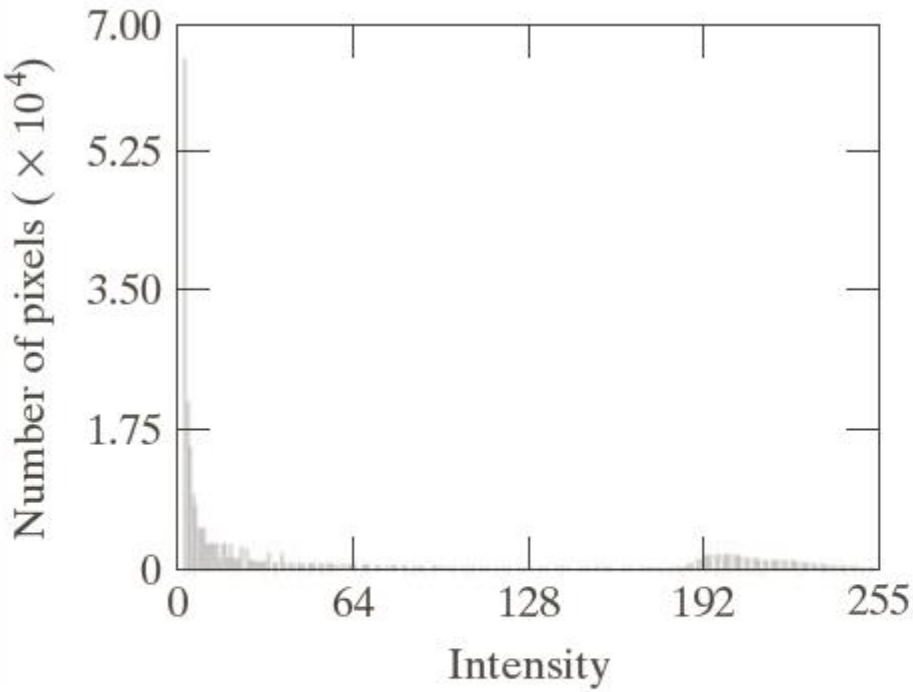


FIGURE 3.22 (a) Histogram of a 3-bit image. (b) Specified histogram. (c) Transformation function obtained from the specified histogram. (d) Result of performing histogram specification. Compare (b) and (d).

Histogram Matching Example 2

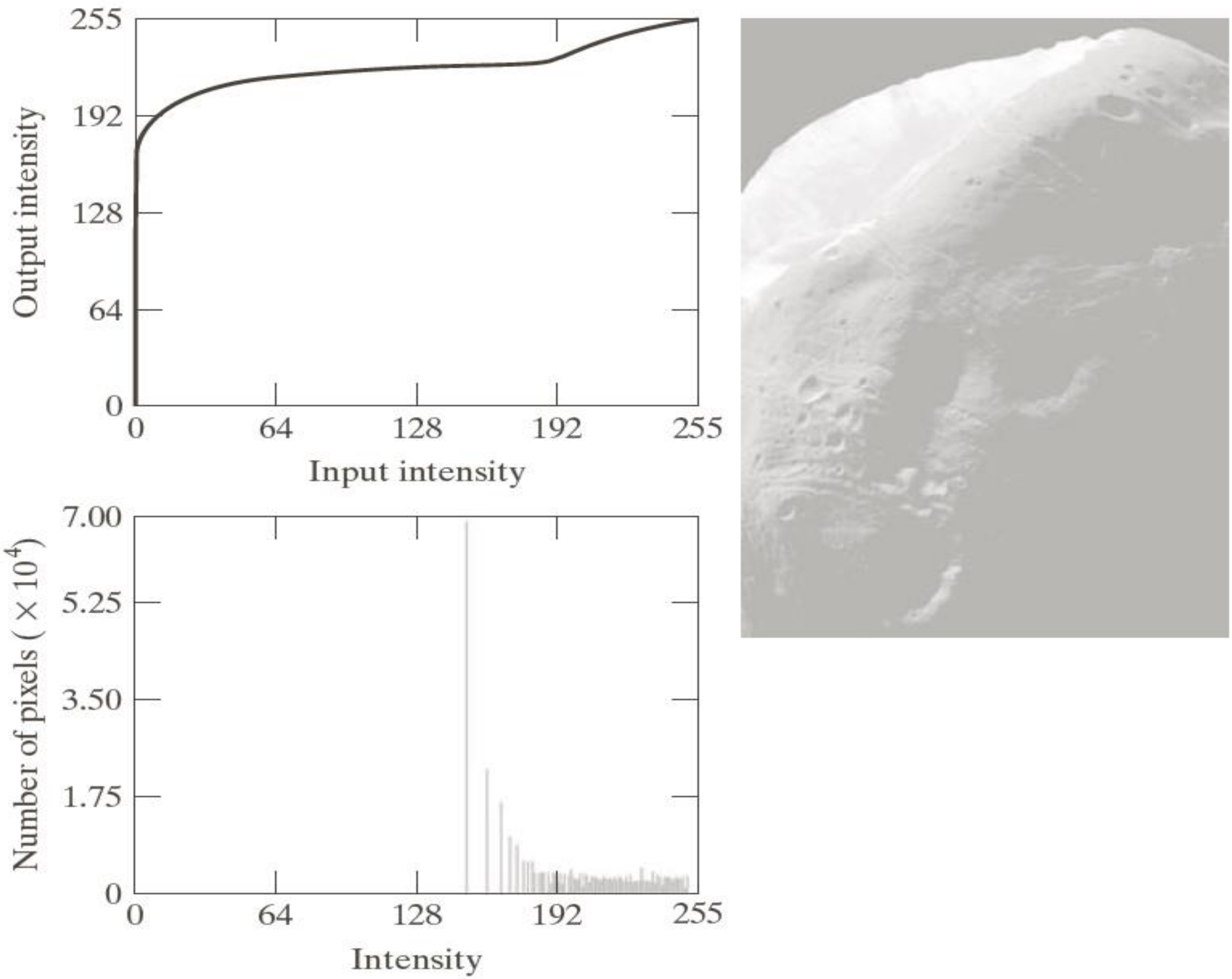




a b

FIGURE 3.23 (a) Image of the Mars moon Phobos taken by NASA's *Mars Global Surveyor*. (b) Histogram. (Original image courtesy of NASA.)

Histogram Matching Example 2

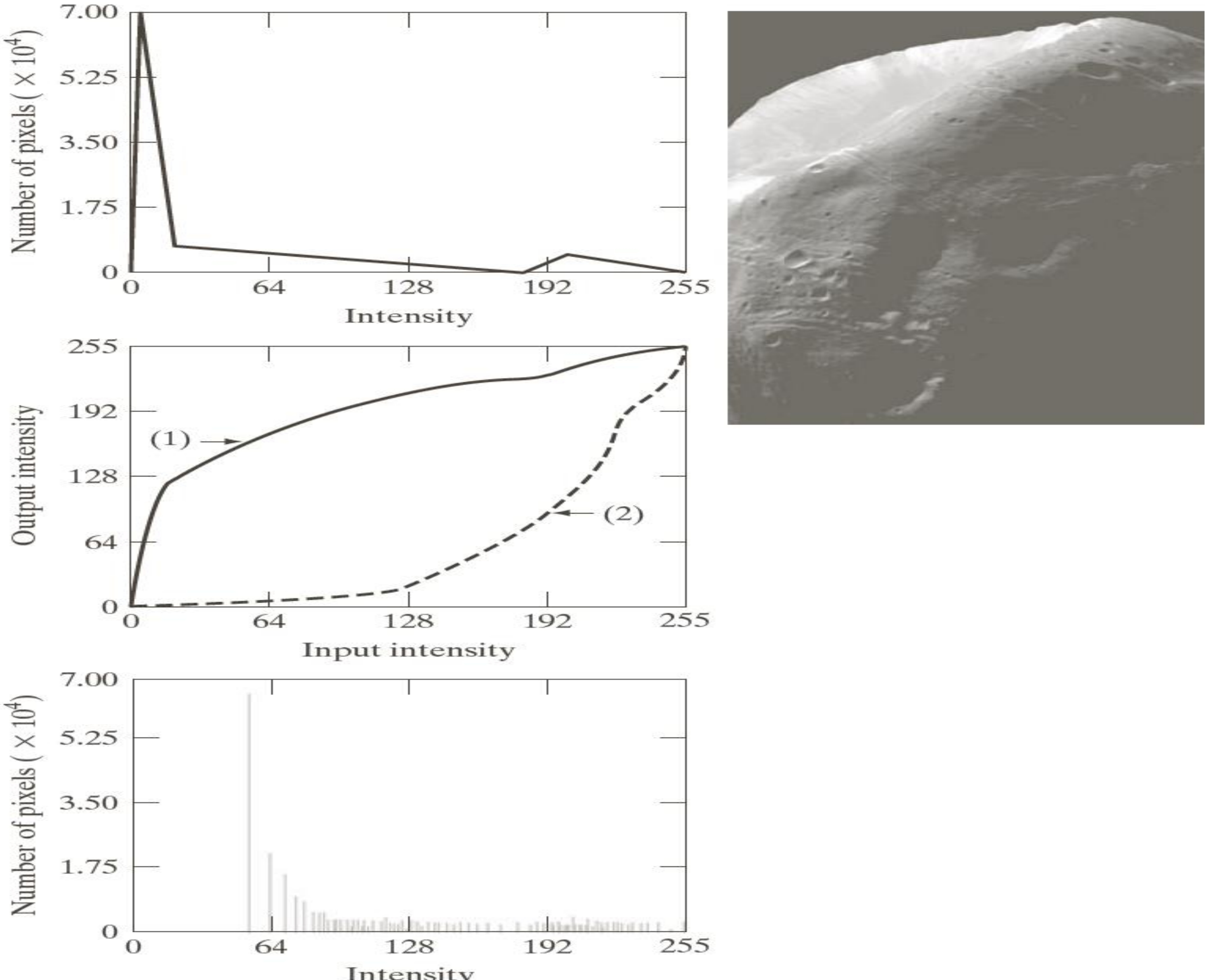


a b
c

FIGURE 3.24
(a) Transformation function for histogram equalization.
(b) Histogram-equalized image (note the washed-out appearance).
(c) Histogram of (b).

Department of Cognitive Science
Xiamen University

Histogram Matching Example 2



a c
b
d

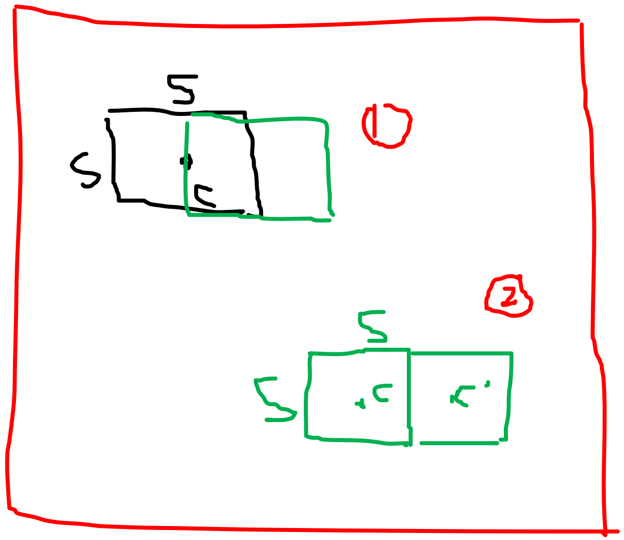
FIGURE 3.25
(a) Specified histogram.
(b) Transformations.
(c) Enhanced image using mappings from curve (2).
(d) Histogram of (c).

Department of Cognitive Science
Xiamen University

3.2.3 Local Histogram Processing

1 Define square or rectangle around each pixel. Move center of window across the image.

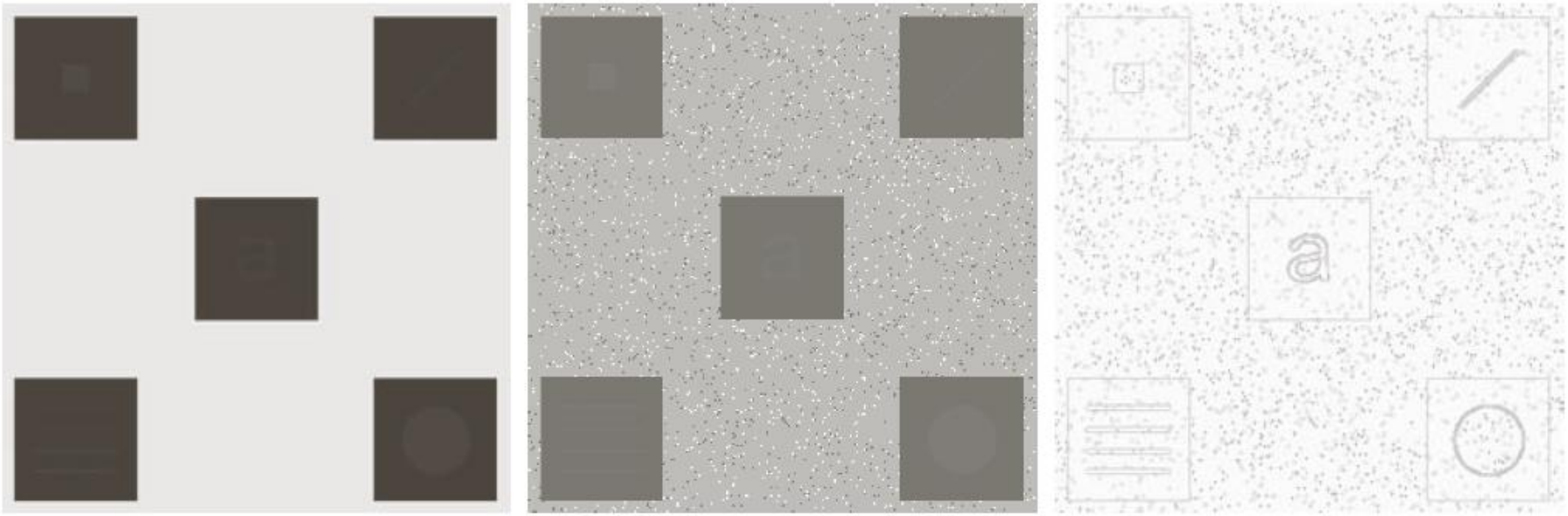
2 At each location, compute histogram in the window. Then do histogram equalization.



blocky

Department of Cognitive Science
Xiamen University

Local Histogram Processing Example



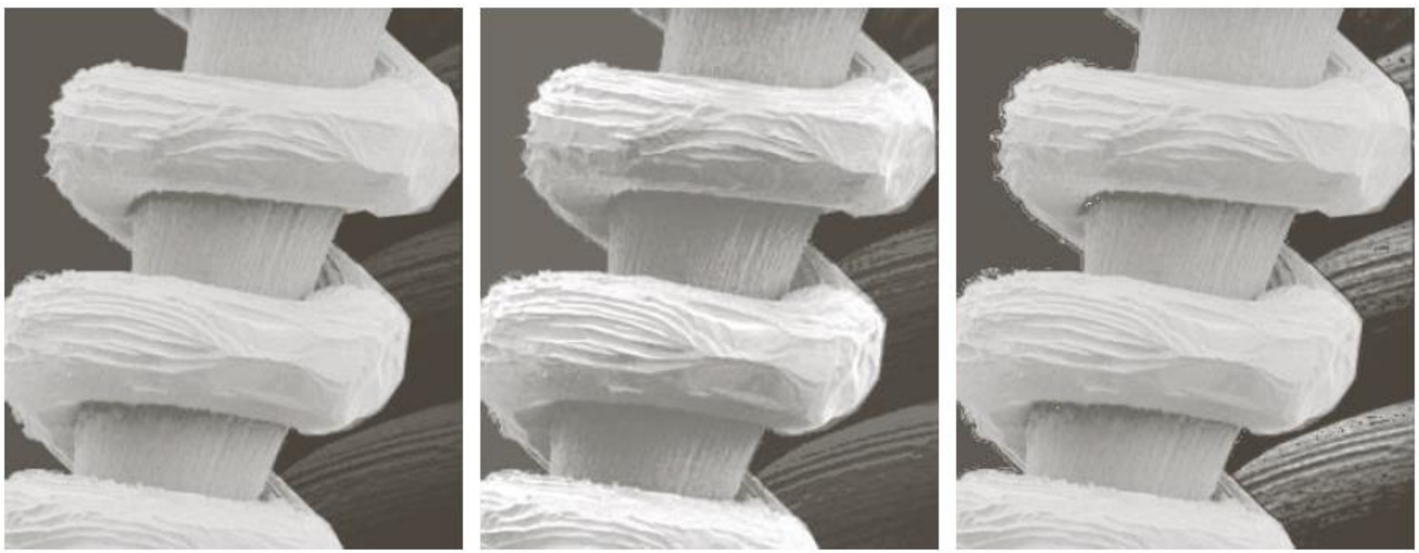
a b c

FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3 × 3.

Histogram statistics for Image Enhancement:

Mean: average intensity

Variance: a measure of average intensity



a b c

FIGURE 3.27 (a) SEM image of a tungsten filament magnified approximately 130×. (b) Result of global histogram equalization. (c) Image enhanced using local histogram statistics. (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)